Autonomous Flights Inside Narrow Tunnels

Luqi Wang, Yan Ning, Hongming Chen, *Graduate Student Member, IEEE*, Peize Liu, Yang Xu, *Graduate Student Member, IEEE*, Hao Xu, Ximin Lyu, and Shaojie Shen

***Abstract*—Multirotors are usually desired to enter confined narrow tunnels that are barely accessible to humans in various applications including inspection, search and rescue, and so on. This task is extremely challenging since the lack of geometric features and illuminations, together with the limited field of view, cause problems in perception; the restricted space and significant ego airflow disturbances induce control issues. This article introduces an autonomous aerial system designed for navigation through tunnels as narrow as 0.5 m in diameter. The real-time and online system includes a virtual omni-directional perception module tailored for the mission and a novel motion planner that incorporates perception and ego airflow disturbance factors modeled using camera projections and computational fluid dynamics analyses, respectively. Extensive flight experiments on a custom-designed quadrotor are conducted in multiple realistic narrow tunnels to validate the superior performance of the system, even over human pilots, proving its potential for real applications. In addition, a deployment pipeline on other multirotor platforms is outlined and open-source packages are provided for future developments.**

***Index Terms*—Aerial systems, applications, autonomous vehicle navigation, field robots, motion and path planning.**

I. Introduction

In recent years, micro aerial vehicles (MAVs), particularly multirotors, have been widely adopted in various types of applications, including inspection [1], search and rescue [2], and surveillance [3], on account of their agility and compact size. These attributes enable multirotors to access confined and narrow spaces that are barely accessible to humans and ground vehicles during those missions, especially in unstructured or indoor environments.

Received 20 November 2024; accepted 31 January 2025. Date of publication 6 March 2025; date of current version 28 March 2025. This work was supported in part by the HKUST-DJI Joint Innovation Laboratory, in part by the HKUST Postgraduate Studentship, in part by the Hong Kong Center for Construction Robotics (InnoHK center supported by Hong Kong ITC), and in part by the Guangdong-Hongkong-Macao Joint Research of Science and Technology Planning Funding from Guangdong Province under Grant 2023A0505010019. This article was recommended for publication by Associate Editor A. Suarez and Editor M. Schwager upon evaluation of the reviewers' comments. *(Corresponding authors: Luqi Wang; Hao Xu; Ximin Lyu.)*

Luqi Wang, Yan Ning, Peize Liu, Yang Xu, and Shaojie Shen are with the Department of Electronic and Computer Engineering, Hong Kong University of Science and Technology, Hong Kong, China (e-mail: lwangax@connect.ust.hk; yningaa@connect.ust.hk; peize.liu@connect.ust.hk; yxuew@connect.ust.hk; eeshaojie@ust.hk).

Hongming Chen and Ximin Lyu are with the School of Intelligent Systems Engineering, SunYat-sen University, Guangzhou 510275, China (e-mail: chenhm223@mail2.sysu.edu.cn; lvxm6@mail.sysu.edu.cn).

Hao Xu is with the Institute of Unmanned Systems, Beihang University, Beijing 100191, China (e-mail: xuhao3e8@buaa.edu.cn).

This article has supplementary downloadable material available at https://doi.org/10.1109/TRO.2025.3548525, provided by the authors.

Digital Object Identifier 10.1109/TRO.2025.3548525

Despite the theoretical capability to navigate in narrow spaces, multirotors face significant perception and control challenges within such environments, which have barely been addressed in previous research. In this article, we focus on one of the most common and desirable yet extremely challenging scenarios: multirotor flights in narrow tunnel-like structures, such as drainage and ventilation conduits or other types of pipelines. As highlighted in previous studies [4], [5], [6], navigating these narrow tunnels presents substantial difficulties in terms of perception and control.

1) In addition to the absence of a global positioning system, the lack of geometric features and external illuminations, as well as the narrow space at corners challenges the perception system with a restricted field of view (FoV).

2) The strong and complicated ego airflow disturbances intensively challenge the control system of multirotors in these confined and restricted spaces, where the disturbances from the complex and chaotic aerodynamic proximity effects can hardly be modeled with simple equations.

Generalized aerial systems using state-of-the-art motion planning methods, for instance [7], are designed for navigating wider indoor or outdoor areas, rather than the confined narrow tunnels. They are tested to be impracticable in such narrow areas, as indicated in [6]. Therefore, a customized aerial platform with proper navigation method is required to conquer the perception and control difficulties and navigate the narrow tunnels. Although the system proposed in [6] offers a potential strategy for navigating certain narrow tunnels, it is primarily designed for near-horizontal tunnels with minor curvatures and unchanged cross-sections that are often overly idealized. In addition, the system operates at a constant speed without considering variations in tunnel shapes, and the complicated speed selection procedure through numerous flights takes a large amount of labor effort, making deployment difficult.

To conquer the aforementioned challenges, we develop an autonomous aerial system distinct from [6] to handle more complex and general scenarios. The newly developed system features a virtual omnidirectional perception module and a novel perception-and-disturbance-aware planning algorithm to navigate through narrow tunnels of varying cross-sectional shapes and sizes. It can also accommodate different directions and corner geometries, which covers most real-world scenarios. The virtual omnidirectional perception module is capable of sensing the surrounding environment forward, upward, and downward, and can achieve omnidirectional perception through yaw movements of the quadrotor. It can be functional in tunnels with any slope while maintaining a compact and lightweight

1941-0468 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Technion Israel Institute of Technology. Downloaded on September 11,2026 at 02:01:38 UTC from IEEE Xplore. Restrictions apply.

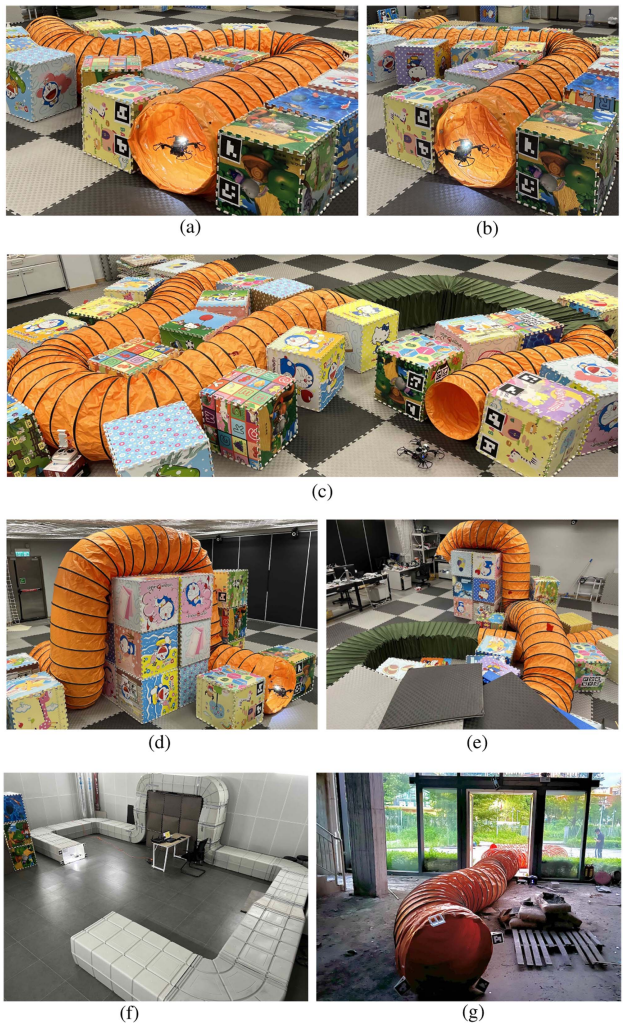


Fig. 1. Narrow tunnels to test the proposed autonomous aerial system. (a) 2-D tunnel case 1. (b) 2-D tunnel case 2. (c) 2-D tunnel case 3 with different cross-section shapes. (d) 3-D tunnel case 1. (e) 3-D tunnel case 2 with different cross-section shapes. (f) Rigid vent pipe with various cross-section sizes and sharp turns. (g) Vent pipe on a construction site.

design to ensure traversability and sufficient power margin. In addition, a novel perception-and-disturbance-aware tunnel planning framework, which utilizes a novel perception and ego airflow disturbance model is proposed. The planner, along with the virtual omnidirectional perception module, is integrated into a quadrotor platform and tested in various narrow tunnels to validate the practicability of the entire system. All the algorithms and modules operate online in real-time to ensure practicability. Finally, the systematic framework is distilled into a comprehensive pipeline for migration to other multirotor platforms for further deployment.

The contributions of this article are summarized as follows.

1) A virtual omnidirectional perception module comprising state estimation, mapping, and tunnel center waypoint extraction, tailored for narrow tunnel flights;
2) A novel perception-and-disturbance-aware motion planning framework that enables a quadrotor to navigate safely and smoothly through narrow tunnels of various cross-sectional shapes and in arbitrary directions;
3) A complete real-time and online narrow tunnel autonomous flight system, which includes the integration of the tailored perception, planning, as well as control and other necessary modules, on a customized quadrotor platform with compact design;
4) Extensive flight experiments and comparison to validate the entire system in real and challenging narrow tunnel scenarios, as shown in Fig. 1;
5) A complete pipeline to extend the developed framework to other multirotor platforms, accommodating different multirotor sizes and more tunnel cross-section shapes. In addition, all components in the system are released as open-source packages for future developments.

To the best of the authors' knowledge, this is the first known autonomous quadrotor system capable of flying through real tunnels in arbitrary directions and as narrow as 0.5 m in cross-section diameter, which outperforms human pilots.

The rest of this article is organized as follows. In Section II, we introduce related work about multirotor narrow tunnel flights. In Section III, we provide an overview of our autonomous system and the tunnel flight workflow. Section IV gives details of the virtual omnidirectional perception module, including state estimation, mapping, and tunnel center waypoint extraction. In Section V, we formulate the perception and ego airflow disturbance models, which facilitate our perception-and-disturbance-aware planning approach described in Section VI. Section VII describes the experiment setup in multiple scenarios and provides results. In Section VIII, we discuss the extendability and possible future improvement of the system. Finally, Section IX concludes this article.

II. Related Work

A. Ego Airflow Disturbances in Narrow Tunnels

Ego airflow disturbances strongly impact rotorcraft flight dynamics in narrow tunnels. These disturbances are primarily due to proximity effects, which are more significant and complex within narrow tunnels compared to open areas. These effects have been extensively studied by researchers for several decades. In general, proximity effects can be categorized into ground, ceiling, and wall effects. The ground effect model, initially proposed by Betz [8] and Chesseman and Bennett [9], revealed that when a rotorcraft flies near ground, additional lift force will be induced. The model is widely adopted by researchers and is analyzed on small scale rotorcrafts in recent years [10], [11], [12], [13]. This well-documented model also provides operational guidelines for real applications [14], [15]. Similarly, the ceiling effect model has also been developed and analyzed for years [13], [16], which reveals that addition lift force will also be generated to pull the rotorcraft up when it flies under a ceiling. The model has also been integrated into flights [17] to enhance safety and control performance. Another proximity effect, the wall effect, characterized by induced forces and torque pushing the rotorcraft toward nearby walls, has also been analyzed extensively [11], [13], [18]. However, a concise wall

Authorized licensed use limited to: Technion Israel Institute of Technology. Downloaded on September 11,2026 at 02:01:38 UTC from IEEE Xplore. Restrictions apply.

effect model as simple as the ground effect model has yet to be established due to its complexity. The disturbances generated from the proximity effects disrupt the control of flights, and can pose significant safety risks. Although in a broader area, the large disturbances near the obstacles can be modeled or avoided [19], in a confined narrow tunnel, the intensified disturbances become more complicated, invalidating the proximity effect models, and there is no space to avoid the high-disturbance zones. Consequently, navigating narrow tunnels remains a formidable challenge.

B. Impact of Planning on Perception and Control in Narrow Tunnel Flights

Planning acts as a bridge between perception and control, impacting both during autonomous flights. In narrow tunnels, maintaining a safety margin by aligning the planned trajectory with the centerline is essential due to significant airflow disturbances and limited maneuvering space. In this context, speed profile planning becomes critical. High-speed flights are usually considered to pose significant challenges for both perception and control systems. High velocities induce issues such as severe motion blur and substantial parallax, complicating stable feature extraction and tracking [20]. Moreover, a low-latency computation required in high-speed flights is also hard to achieve [21]. In addition, control performance generally degrades with increasing flight speed due to modeling inaccuracies and the physical limitations of multirotors. As indicated in [22] and [23], higher flight speeds lead to larger control errors, increasing the risk of collisions near obstacles. As a result, strategies such as adjusting safety margins for flights at different speeds [23] or adjusting speed based on available space [24] are proposed to mitigate the risk. However, previous studies have shown that slow flight speeds near obstacles can intensify ego airflow disturbances caused by proximity effects, potentially causing larger control tracking errors than those at higher speeds [19]. Thus, flying at slow speeds and neglecting proximity effects can also be risky during flights. In narrow tunnels, where perception is difficult and ego airflow disturbances are extremely severe, it is crucial to maintain appropriate flight speeds by considering both factors [6].

C. Multirotor Systems in Tunnel-Like Confined Areas

Although multirotor flights in cluttered environments have been extensively studied [7], [25], [26], [27], the exploration of multirotor capabilities in tunnel-like confined areas remains limited due to intense challenges. As highlighted in prior studies [4], [28], difficulties in state estimation within tunnels arise due to the lack of light and geometry features. In addition, control issues are exacerbated by significant ego airflow disturbances resulting from proximity effects [5]. Previous attempts to develop tunnel navigation systems have been inadequate. For instance, a minimum-time tunnel-following navigation was proposed in [29] to follow a tunnel, while the authors overlooked the critical perception and ego airflow disturbance issues and relied solely on simulation results, making it impractical for real-world application. Another tunnel navigation method with an integrated system was proposed in [30]. However, the proposed system only follows horizontal large tunnels without addressing key perception and control issues, making it ineffective in narrow tunnels as small as 0.5 m in diameter. Although a quadrotor system in [6] was proposed with the capability of navigating narrow tunnels, as mentioned in Section I, the system is not complete for applications in generalized narrow tunnels. It is limited to nearly horizontal tunnels with minimal curvature and consistent cross-sections, and it utilizes a constant speed planning approach that is not suitable in more general cases.

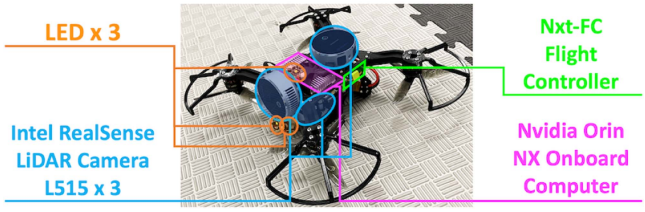


Fig. 2. Customized quadrotor platform for narrow tunnel flights.

III. System Overview

A. Quadrotor Platform Design

Given the impracticality of installing external localization systems in all potential tunnel environments during real-world deployment, an onboard perception system is essential for autonomous tunnel navigation. As mentioned in [6], the tunnel's internal environment typically lacks geometry features and external illumination, making visual-inertial odometry together with supplementary LED lighting the most viable state estimation strategy at present. In addition, the significant variations in illumination conditions at tunnel entrances and exits can severely disrupt vision-based depth estimation, making a relatively light-insensitive LiDAR a practical solution for the perception module. Consequently, we employ LiDAR-based RGB-D cameras as our primary perception sensor.

To ensure safe navigation through narrow tunnels with limited maneuvering space, it is crucial to maintain a flight path along the tunnel's centerline to gain maximum possible safety margins. Furthermore, tunnels may extend in various directions in practice, including vertically, the quadrotor must possess omnidirectional perception capabilities to safely navigate such confined spaces. Given that only yaw motion can be utilized for turning, the perception module still requires sensing in three directions: front, upward, and downward, as the minimal combination to achieve virtual omnidirectional perception.

Therefore, we design a customized quadrotor platform equipped with three RGB-D camera units and three LEDs, oriented in the aforementioned directions. In addition, the platform is designed with a compact and light structure, as shown in Fig. 2. It utilizes 5-in propellers, features a diameter of 40 cm, a wheelbase of 25 cm, and a total weight of just 1.085 kg, despite carrying more equipment.

Authorized licensed use limited to: Technion Israel Institute of Technology. Downloaded on September 11,2026 at 02:01:38 UTC from IEEE Xplore. Restrictions apply.

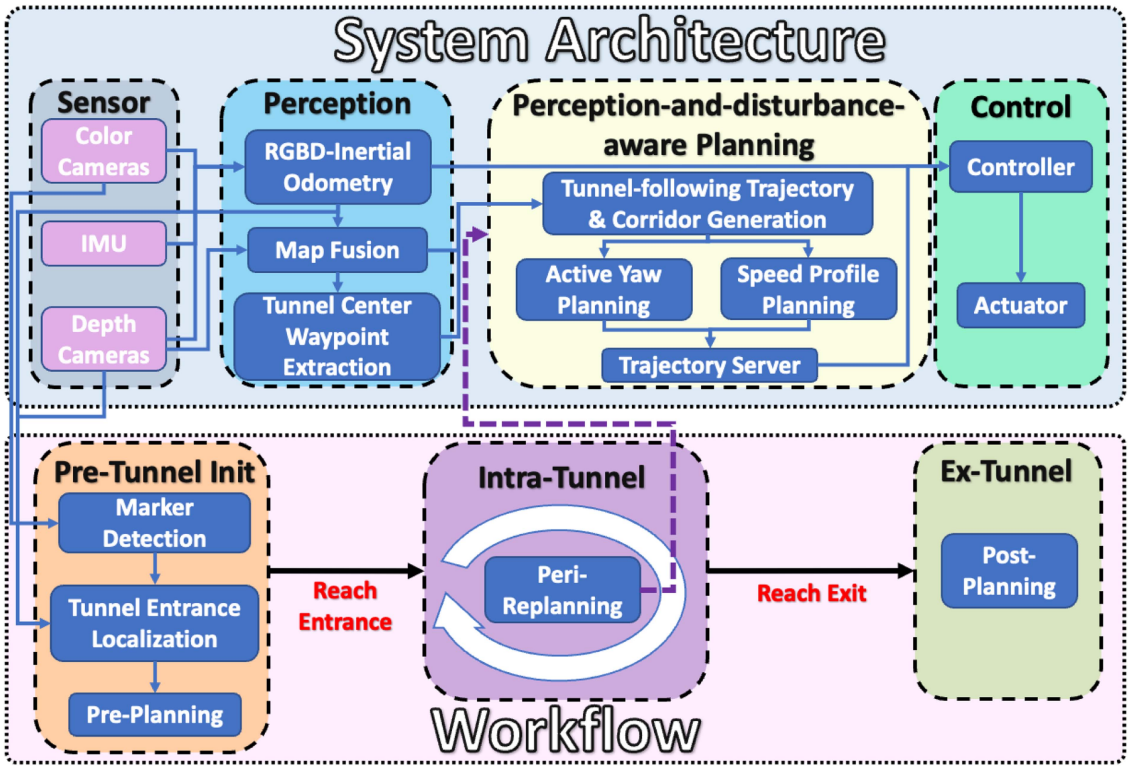


Fig. 3. System architecture and the workflow for narrow tunnel flights. The perception module adopts the color and depth images, as well as IMU data to produce RGBD-inertial odometry for state estimation and then performs map fusion. The tunnel center waypoints extracted from the mapping result facilitate the perception-and-disturbance-aware planning, which generates flight trajectories, yaw trajectories, and speed profiles. The results are then utilized to generate control commands and executed by the motors. The workflow consists of pretunnel initialization, which performs preplanning to the localized tunnel entrance according to the detected marker, intratunnel peri-replanning and extunnel postplanning.

B. System Architecture

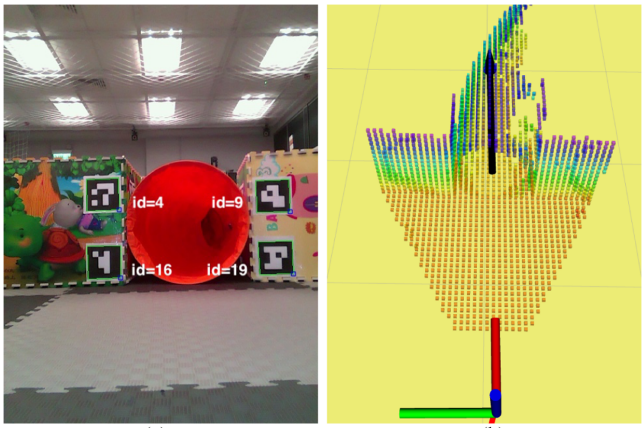
To facilitate narrow tunnel flights using the customized quadrotor platform, a comprehensive software framework shown in Fig. 3 is integrated on the onboard computer. The framework consists of perception, planning, and high-level control. To fully utilize the color and depth images from multiple camera modules, we developed an RGBD-inertial state estimation system that fuses these images with IMU data. (see Section IV-A). The fused RGBD-inertial odometries and depth images are then fed to the map fusion module to update occupancy map and the corresponding Euclidean distance field (EDF) (see Section IV-B). The updated EDF then facilitates the tunnel center waypoint extraction (see Section IV-C). The perception-and-disturbance-aware planning module leverages a perception (see Section V-A) and an ego airflow disturbance model (see Section V-B) to generate trajectories from the extracted waypoints. The planning module performs flight trajectory and corridor generation (see Section VI-A), active yaw planning (see Section VI-B) and speed profile planning (see Section VI-C). Subsequently, the position, velocity, and acceleration commands are generated from the planned results and sent to the high-level controller. Then, the high-level controller generates corresponding attitude and thrust commands, which are fed to the flight controller to produce the necessary low-level commands executed by the motors.

C. Tunnel Flight Workflow

As the quadrotor needs to enter and exit a narrow tunnel safely and smoothly, a workflow containing pretunnel initialization, intratunnel replanning, and extunnel postplanning, shown in Fig. 3, is developed.

1) *Pretunnel initialization:* Before the flight, four ArUco markers are symmetrically placed at the tunnel entrance to aid in entrance localization, as shown in Fig. 4. The quadrotor utilizes the captured color and depth images, along with the estimated camera poses to detect the markers and calculate their global 3-D positions. Then, a random sample consensus algorithm is applied to the marker positions to reject outliers and ensure accuracy. The position and direction of the tunnel entrance are determined by averaging the four filtered marker positions and calculating the normal of the least-squares error plane constructed from these positions, respectively. This estimated tunnel entrance pose allows the quadrotor to generate a smooth B-spline trajectory toward it, completing the pretunnel initialization phase;
2) *Intratunnel replanning:* Upon reaching the tunnel entrance, the quadrotor switches to the intratunnel state. It starts the mapping module and extract the tunnel shape to perform perception, while initiates

Authorized licensed use limited to: Technion Israel Institute of Technology. Downloaded on September 11,2026 at 02:01:38 UTC from IEEE Xplore. Restrictions apply.



(a) (b)

Fig. 4. Illustration of entrance localization of the tunnel shown in Fig. 1(c). The color code indicates the height; the axis indicates the pose of the quadrotor and the black arrow indicates the estimated tunnel entrance pose. (a) Detected ArUco markers and the corresponding marker IDs. (b) Visualization of the detected entrance and the voxel map.

an ego-airflow-disturbance-and-perception-aware replanning process that continues throughout the tunnel traversal. Details on perception and planning will be elaborated in Sections IV and VI, respectively.

3) *Extunnel postplanning:* As the quadrotor approaches the tunnel exit, a smooth B-spline trajectory is generated to safely decelerate it during the posttunnel phase. This ensures a smooth transition from the narrow tunnel to the wider external area.

IV. Perception in Narrow Tunnels

The quadrotor must have perception capabilities to navigate narrow tunnels, where no assistive infrastructure is available. In such constrained space, efficient and accurate perception is necessary to ensure flight safety. Therefore, the virtual omnidirectional perception module needs to fully utilize the information from the three RGB-D cameras oriented forward, upward, and downward. In this section, we will elaborate on the details of this module, covering state estimation, mapping, and tunnel center waypoint extraction.

A. State Estimation in Narrow Tunnels

We develop an optimization-based multicamera-IMU state estimator based on our previous works [31], [32]. The estimator processes color and depth images from three individual RGB-D camera modules to produce RGBD-inertial odometries. The system architecture of the state estimator is illustrated in Fig. 5. The estimator comprises three main components: parallel front ends, a front end coordinator, and a back end optimizer.

1) Parallel Front End: Each visual front end independently processes color and depth images from its respective camera, performing feature detection, tracking, and outlier rejection. Concurrently, the IMU front end conducts preintegration of inertial data, providing high-frequency odometry updates based on the latest optimized states.

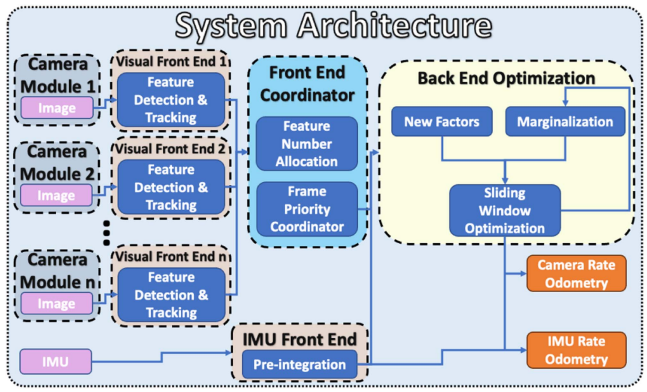


Fig. 5. System architecture of the state estimator in narrow tunnels. The estimator processes color and depth images from individual camera modules to produce RGBD-inertial odometries. It includes multiple parallel front ends, a front end coordinator, and an optimization back-end. In practice, it utilizes three RGB-D camera modules.

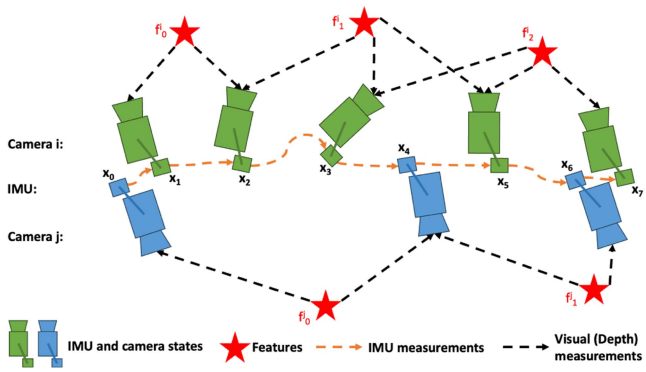


Fig. 6. Illustration of the sliding window for optimization in the state estimation.

2) Front End Coordinator: Given the different perspectives in the tunnel, the feature qualities vary across cameras. In order to efficiently utilize the features and computation resources, the front end coordinator module is introduced to dynamically manage the feature allocation and frame prioritization across all camera front ends. By utilizing a dynamic feature number allocation strategy, it adjusts the number of features each camera processes based on the quality and quantity of features detected in the previous frame. This strategy helps maintain consistent feature tracking across different camera views.

3) Back End Optimizer: The back end of the state estimator employs a sliding window optimization approach, integrating visual, depth, and IMU measurements to estimate the state vector, as illustrated in Fig. 6. This state vector includes the position, velocity, orientation, and sensor biases of the IMU. By chronologically ordering asynchronous frames within the sliding window, the optimizer ensures coherent and accurate state estimation. The optimization objective combines prior factors, IMU propagation factors, and visual-depth factors to refine the state estimates continuously.

By leveraging the three RGB-D camera modules and an efficient coordination mechanism, the optimization-based state estimation approach provides robust RGBD-inertial odometries

Authorized licensed use limited to: Technion Israel Institute of Technology. Downloaded on September 11,2026 at 02:01:38 UTC from IEEE Xplore. Restrictions apply.

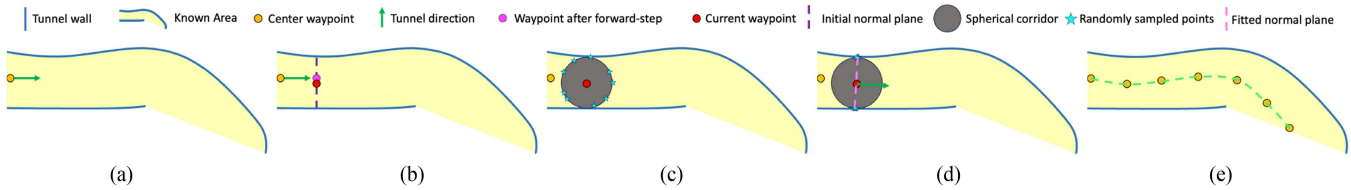


Fig. 7. Illustrations of tunnel center waypoint extraction using EDF. The illustration is in 2-D for a clear view. (a) Initial center waypoint and direction. (b) Step forward and gradient ascend in the normal plane according to the EDF value. (c) Randomly sample points on the spherical corridor. (d) Gradient descend the sampled points on the sphere according to the EDF value and fit the new tunnel direction. (e) Iterate (b) to (d) to obtain the final tunnel center waypoint set.

during the navigation through narrow tunnels. For further details about the state estimator, we suggest the readers refer to [33].

B. Mapping in Narrow Tunnels

To perform mapping in narrow tunnels in arbitrary directions, it is also essential to fully utilize the depth information. We developed a parallel and asynchronous mapping module based on [34]. This module processes depth images from three cameras and RGBD-inertial odometry data from the state estimation module to construct occupancy and EDF maps inside tunnels. The projection and ray-casting procedures for each camera are assigned to separate threads, which asynchronously update the occupancy map with the corresponding camera odometries. Meanwhile, the EDF map, which stores the minimum Euclidean distance of each voxel to the nearest obstacle, is updated synchronously. This strategy efficiently manages the asynchronous depth images from the three cameras, enabling omnidirectional mapping with yaw movements in narrow tunnels.

C. Tunnel Center Waypoint Extraction

To ensure safety and control within narrow tunnels, it is crucial to maintain flight paths along the centerline, keeping as far from the walls as possible. This minimizes proximity effects that can cause strong external forces and unpredictable disturbances, reducing control difficulties and enhancing safety [10], [11], [19]. Therefore, extracting tunnel center waypoints, which represent the geometric details of the tunnel centerline, is essential for tunnel flight operations.

The procedure for the tunnel center waypoint extraction is illustrated in Fig. 7 and detailed in Algorithm 1. It utilizes the EDF map generated by the mapping module to extract the center waypoints. By providing the start point p_0 and direction dir_0, the algorithm searches along the tunnel with a constant step length S until the total planned distance d_p reaches a preset plan range R_p, or the searching point p reaches an unknown or impassable area. Specifically, traversal is considered invalid if twice the EDF value at p is smaller than the quadrotor diameter D_q or larger than the maximum tunnel diameter $D_{\max}$. The maximum allowable EDF value $d_{\max}$ is initialized to the max radius of the tunnel and the search point p is initialized to the maximum EDF value position in the plane with the normal direction dir_0 and a distance S ahead of the initial point p_0. Using the gradient ascent method and the maximum allowable EDF value $d_{\max}$, a proper initialization can be achieved. During the gradient ascent, if the initial point has an EDF value larger than the maximum allowable EDF value, a gradient descent is first conducted until the EDF value reaches the maximum allowable limit. This is particularly useful in the prediction stage when only part of the tunnel wall is observed in the local map, preventing the point from ascending unreasonably away from one side of the wall. During each iteration of the search, at a new search point p, a

Algorithm 1: Tunnel Center Waypoint Extraction.

Notation: Start point p_0, Start direction dir_0, Tunnel center waypoints $\mathcal{W}$, Point p, Quadrotor dimension D_q, Maximum tunnel dimension $D_{\max}$, Search step length S, Search direction change threshold $DIR_{\min}$, EDF value at a position $edf(\cdot)$, Point set $\mathcal{P}$, Plan distance d_p, Plan range R_p
Input: p_0, dir_0
Output: $\mathcal{W}$
Initialize :
$predict_dir \leftarrow false$
$d_p \leftarrow 0$
$dir^- \leftarrow dir_0$
$dir^+ \leftarrow dir_0$
$d_{\max} \leftarrow 0.5 \cdot D_{\max}$
$\mathcal{W}$.push_back(p_0)
$p \leftarrow GradientAscend(p_0 + S \cdot dir^+, dir^+, d_{\max})$
while $d_p \leq R_p$ && *is_known*(p) && $D_q \leq 2 \cdot edf(p) \leq D_{\max}$ **do**
  $\mathcal{W}$.push_back(p)
  $\mathcal{P} \leftarrow SphereRandomSample(p)$
  for each $p_i \in \mathcal{P}$ **do**
  $p_i \leftarrow SphereGradientDescend(p_i)$
  end for
  $dir^- \leftarrow dir^+$
  $dir^+ \leftarrow PlaneFit(\mathcal{P}, dir^-).normal()$
  if $predict_dir$ **then**
  $dir^+ \leftarrow dir^-$
  else
  if $dir^- \cdot dir^+ \leq DIR_{\min}$ **then**
  $predict_dir \leftarrow true$
  $d_{\max} \leftarrow edf(p)$
  end if
  end if
  $p \leftarrow GradientAscend(p + S \cdot dir^+, dir^+, d_{\max})$
  $d_p \leftarrow d_p + dist(p, \mathcal{W}.back())$
end while
return $\mathcal{W}$

Authorized licensed use limited to: Technion Israel Institute of Technology. Downloaded on September 11,2026 at 02:01:38 UTC from IEEE Xplore. Restrictions apply.

bunch of random points are sampled on the spherical corridor, which is a sphere centered at p with a radius equal to the EDF value at p. Each sampled point p_i undergoes gradient descent to the position on the sphere closest to the tunnel wall, minimizing the EDF value. The tunnel direction dir^+ is obtained through least-squares plane fitting of the descended points. Finally, a gradient ascent is performed on the plane with a normal direction dir^+ and a distance S ahead of the current point to complete the loop. If the change in tunnel direction compared with the previous direction dir^- is significant, it usually indicates incomplete tunnel observation. In such cases, predictions should be made from the current point along the previous direction, assuming the tunnel dimensions remain unchanged. The waypoints from each iteration form the tunnel center waypoint set $\mathcal{W}$, concluding the center waypoint extraction.

V. Model Formulation of Perception and Ego Airflow Disturbance Factors

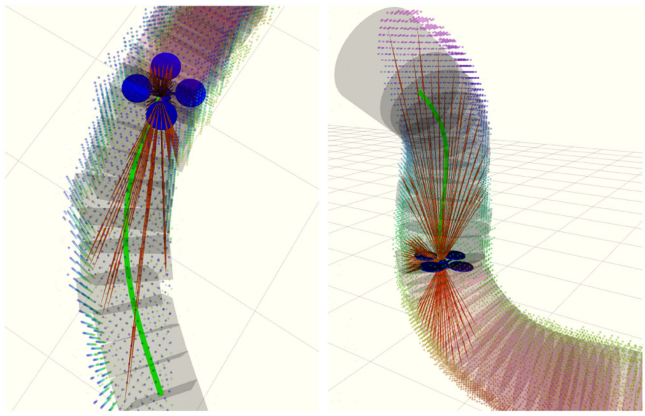
After ensuring the perception capability, motion planning inside narrow tunnels can be discussed. However, prior to introducing the planning methodology, it is important to formulate the perception and ego airflow disturbance models, as these two factors significantly impact performance and safety during flights through narrow tunnels, as indicated in Section I.

As analyzed in [6], the perception and ego airflow disturbance factors are often coupled and exhibit opposing effects on flight performance with changes in speed, necessitating compensation during planning. However, the detailed formulation of these factors remains to be analyzed. Therefore, we first analyze the perception factor related to feature tracking, as its stability is crucial for effective perception, and model the mean optical flow speed. Next, we examine the ego airflow disturbance, which significantly impacts control performance, and model the disturbance level under different flight conditions. With these models in place, perception-and-disturbance-aware planning can be conducted afterward.

A. Perception Factor Model Formulation

The experimental results in [6] demonstrate that as flight speed inside the tunnel increases, the number of tracking features decreases, indicating reduced perception performance. Since a vision-based state estimation system is used, feature tracking occurs on 2-D images. It is more accurate to conclude that the number of tracked features decreases as the optical flow speed on the images increases. Therefore, to ensure stable feature tracking and improve perception quality for safe flights, it is important to minimize the optical flow speed on the images.

Predicting the optical flow speed of future features is typically challenging, but the tunnel scenario simplifies this because the features are located on the tunnel walls. For a 2-D feature point on the image planes of the cameras along the tunnel-following trajectory (will be detailed in Section VI-A1), the corresponding 3-D points on the wall can be determined using ray-casting, as depicted in Fig. 8. For a 3-D feature point located at $\mathbf{p_f}$ in the camera frame, when the camera is moving at a linear velocity $\mathbf{v_{cam}}$ and an angular velocity $\omega_{\mathbf{cam}}$, as illustrated in Fig. 9, the



(a) (b)

Fig. 8. Illustrations of the uniformly sampled feature points on the image planes and the corresponding 3-D points on the extracted tunnel walls in simulations. The green curves represent the tunnel-following trajectory, the transparent boxes and cylinders indicate the estimated tunnel cross-sections, and the red arrows indicates the corresponding 3-D positions of the sample points on the image planes. Note that the quadrotor is equipped with three cameras facing front, top, and down, and only the samples of the current position of the quadrotor is illustrated in each case. (a) Quadrotor is flying in a horizontal rectangular tunnel. (b) Quadrotor is flying upward in a circular tunnel.

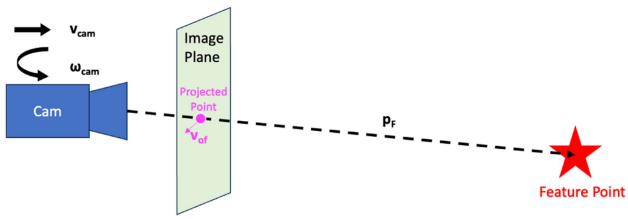


Fig. 9. Illustration of the projection model for a feature point.

optical flow velocity on the image plane $\mathbf{v_{of}}$ can be derived as follows:

$$\mathbf{v_{of}} = \text{proj}\left(\mathbf{K_{cam}} \frac{-\omega_{\mathbf{cam}} \times \mathbf{p_F} - \mathbf{v_{cam}}}{p_{f_z}}\right) \tag{1}$$

where the proj function projects the vector inside the bracket onto the image plane by extracting the x and y elements for the conventional camera frame definition, $\mathbf{K_{cam}}$ is the projection matrix of the camera, and p_{f_z} is the z element of the position of the feature point in the camera frame, which also represents the depth. From the uniform sampling on the camera image plane, the mean optical flow speed can be approximated.

B. Ego Airflow Disturbance Analysis and Modeling

The other crucial factor to analyze is the ego airflow disturbance, as mitigating its effects can enhance control performance. While it might seem that higher flying speeds reduce the impact of airflow disturbances, this intuition mainly applies to horizontal flights. A comprehensive analysis is still necessary to fully understand the effect. Therefore, detailed computational fluid dynamics (CFD) analyses at various angles, flight speeds, and tunnel shapes are conducted to gain a thorough understanding

Authorized licensed use limited to: Technion Israel Institute of Technology. Downloaded on September 11,2026 at 02:01:38 UTC from IEEE Xplore. Restrictions apply.

of airflow disturbances during tunnel flights. Due to the impracticality of performing CFD analyses for every possible scenario, a range of settings representing typical real flight conditions is adopted. The cases encompass speeds from 0 m/s to 3.5 m/s with 0.5 m/s intervals, tunnel pitch direction from -90° (upward) to 90° (downward) with 15° intervals, circular cross-sections with diameters from 0.5 m to 0.9 m with 0.1 m intervals, and rectangular cross-sections with edge lengths from 0.5 m to 0.9 m with 0.1 m intervals, culminating in a total of 2940 cases. In each case, a near-hovering model of the proposed 1.085 kg quadrotor platform with four 5-in propellers is simplified to four 5-in thin fans generating a pressure jump of 210 Pa each. The quadrotor is placed in the middle of a 7-m-long tunnel computation region. Since the quadrotor follows the tunnel centerline, symmetry considerations allow the computational region to be reduced by half, expediting the analyses. In each case, a tetrahedral mesh sized from 0.5 to 4 cm is used, depending on the distance from the quadrotor. A moving frame along the quadrotor is set to simulate the flights at various speeds. The airflow together with the tunnel wall moves backward against the static quadrotor, while both ends of the tunnel maintain atmospheric pressures. The cases are solved using a pressure-based, standard k-epsilon turbulence model and standard wall model using the SIMPLE algorithm in Ansys Fluent.

The backward streamlines from the propellers in selected cases are illustrated in Fig. 10. The airflow varies significantly in different situations, underscoring the importance of comprehensive CFD analyses. Since the primary airflow disturbances are attributable to the turbulent flow generated by the propellers, the disturbance level is formulated as the proportion of the turbulent flow intake relative to the total intake of the propellers. According to the streamline results, this formulation can be approximated by dividing the number of backward streamlines from the propellers by the total number of sampled streamlines, which is 200 for each case in practice. The final disturbance level results for selected cases are depicted in Fig. 11. Generally, the disturbance level decreases when flying faster in a horizontal tunnel. However, this trend does not necessarily hold when the tunnel has a slope. At specific flight speeds and tunnel slopes, for instance as shown in Fig. 10(d)–(f), the quadrotor can easily fly into the turbulent flow it generates, leading to a high disturbance level. This finding contradicts to the simple intuition that faster flights reduce disturbances. To ensure safe flights, these situations should be avoided, necessitating the reduction of disturbance levels as much as possible.

The raw disturbance level data, while derived, cannot be directly used for planning. This is because the flight conditions typically do not match specific CFD cases, and conducting infinite CFD analyses for every scenario is impractical. To address this, a general regression neural network (GRNN) [35] is employed to regress the raw disturbance level data, thereby generating a continuous model. During the training procedure, the raw disturbance level data are firstly divided into training and test sets with a 4:1 ratio. Then, we apply the GRNN on the test data based on training data and find the minimum mean squared error and the corresponding optimal value of the network parameters. Despite the applicability of this continuous model in subsequent planning, a significant drawback of the GRNN is that its computational cost increases with the size of the raw data sample, leading to slower computation as more CFD results are provided. To address this issue, a learning-based regression model acceleration technique is employed, utilizing a multilayer perceptron (MLP) neural network to represent the model. We uniformly sample 1 000 000 points within the domain and compute the disturbance level using the GRNN model. The data are then divided into training, validation, and test sets in a 3:1:1 ratio. The final MLP model achieves a speed improvement of over ten times with a negligible loss of 1e-5 level compared to the GRNN when the raw disturbance level data is in the thousands' level. Information about the MLP models is detailed in Table I.

TABLE I
INFORMATION OF THE MLP NEURAL NETWORK DISTURBANCE MODEL

Model	Circle	Rectangle
Structure	3 layers with 64, 64, 16 hidden sizes	
Loss function	MSE	
Learning rate	$1e-3$ to $1e-6$	
Optimizer	Adam	
Total epoch	50000	70000
Final training loss	$1.1e-5$	$1.8e-5$
Final test loss	$1.3e-5$	$1.9e-5$

VI. PERCEPTION-AND-DISTURBANCE-AWARE PLANNING

With the formulated perception and disturbance models, perception-and-disturbance-aware planning can be executed, leveraging the mapping results and extracted tunnel center waypoints from the perception module. This planning process includes tunnel-following trajectory and corridor generation, perception-aware active yaw planning, and perception-and-disturbance-aware speed profile planning. The planned trajectories and speed profiles are used to generate position, velocity, and acceleration commands, which are then sent to the controller.

A. Tunnel-Following Trajectory and Corridor Generation

To ensure safe flights through narrow tunnels, it is essential to have a collision-free trajectory that follows the tunnel centerline while accounting for system dynamics. In addition, as discussed in Section I, the quality of feature tracking and the level of disturbances, which affect perception and control, vary with tunnel shapes, sizes, and flight speeds. Therefore, identifying the tunnel corridor, including the shape and size of cross-sections, is crucial for subsequent active yaw planning and speed profile planning.

As a result, we propose a tunnel-following trajectory and corridor generation approach that comprises tunnel-following trajectory optimization, cross-section recognition, and trajectory refinement. This integrated approach ensures that the generated trajectories and corridors accurately capture the necessary geometric data of the tunnel while considering smooth system dynamics.

1) Tunnel-Following Trajectory Optimization: With the acquired tunnel center waypoints mentioned in Section IV-C, a

Authorized licensed use limited to: Technion Israel Institute of Technology. Downloaded on September 11,2026 at 02:01:38 UTC from IEEE Xplore. Restrictions apply.

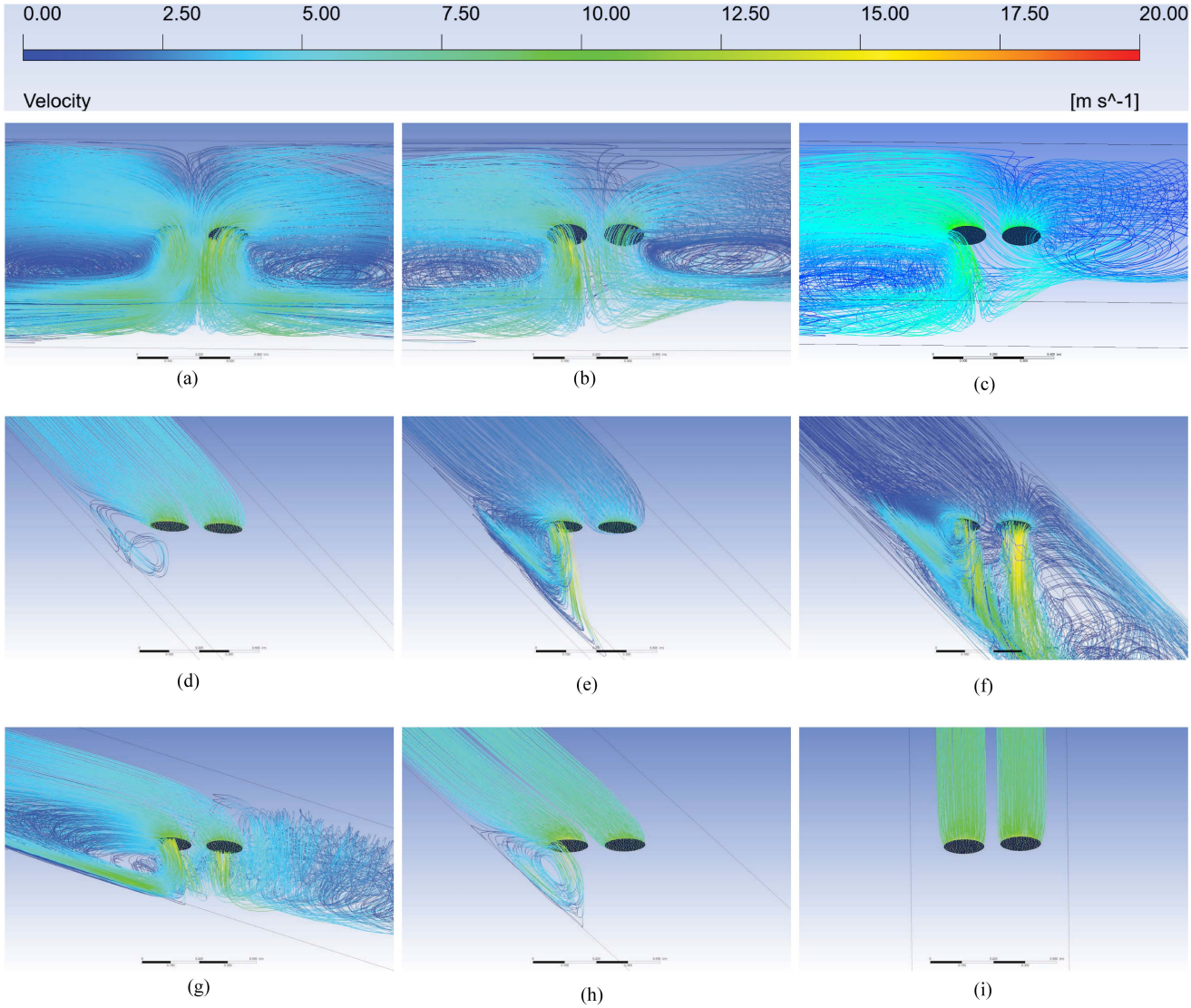


Fig. 10. Backward streamlines from the shadow of the propellers when the quadrotor is flying at different speeds in tunnels of different shapes from the CFD result. The black circles represent the propellers and the color indicates the speed of the flow on the streamlines. Note that the illustrations of half of the space can fully represent the results on account of the symmetry. (a) Hover, horizontal tunnel with 0.6 m edge length square cross-section. (b) 1 m/s toward left, horizontal tunnel with 0.6 m edge length square cross-section. (c) 2 m/s toward left, horizontal tunnel with 0.6 m edge length square cross-section. (d) Hover, 45° slope tunnel with 0.9 m $\times$ 0.5 m rectangular cross-section. (e) 2 m/s toward bottom right, 45° slope tunnel with 0.9 m $\times$ 0.5 m rectangular cross-section. (f) 3.5 m/s toward bottom right, 45° slope tunnel with 0.9 m $\times$ 0.5 m rectangular cross-section. (g) 0.5 m/s toward bottom right, 15° slope tunnel with 0.5 m diameter circular cross-section. (h) 0.5 m/s toward bottom right, 45° slope tunnel with 0.5 m diameter circular cross-section. (i) 0.5 m/s toward bottom right, vertical tunnel with 0.5 m diameter circular cross-section.

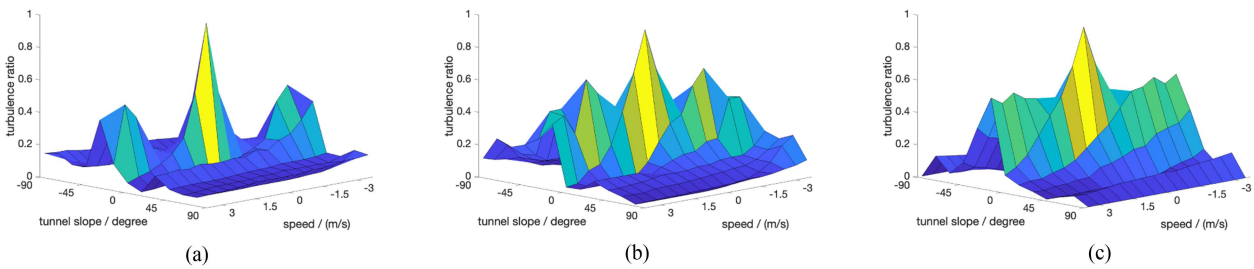


Fig. 11. Disturbance level when the quadrotor is flying at different speeds in tunnels with different slopes and cross-sections. (a) 0.6 m edge length square cross-section. (b) 0.5 m $\times$ 0.8 m rectangular cross-section. (c) 0.5 m diameter circular cross-section.

Authorized licensed use limited to: Technion Israel Institute of Technology. Downloaded on September 11,2026 at 02:01:38 UTC from IEEE Xplore. Restrictions apply.

uniform B-spline through them can be generated to represent the initial estimation of the tunnel centerline. However, due to perception noise, this initial centerline is typically jerky and may deviate from the true centerline, making it unsuitable for smooth flight. In addition, the cross-sections on the jerky initial centerline can significantly differ from the true shape due to the inaccurate directions. Therefore, we introduce a trajectory optimization method that leverages initial waypoints and EDF information from the fused map, while considering system dynamics, to determine the tunnel-following trajectory. This trajectory, represented by a B-spline curve, also facilitates subsequent smooth cross-section recognition.

To enhance the quality of the B-spline trajectory, we propose a cost function $f_{@cl}$ for its evaluation and optimization. This function is a weighted sum of the smoothness cost f_s, the waypoint constraint cost f_w, the interval constraint cost f_i, the distance constraint cost f_d, and the end state constraint costs f_e as follows:

$$f_{cl} = \lambda_s f_s + \lambda_w f_w + \lambda_i f_i + \lambda_d f_d + \lambda_e f_e. \tag{2}$$

Taking account of the smooth dynamics of the multirotor system, a third-order elastic band cost [36], [37] along the trajectory is adopted as follows:

$$f_s = \sum_{i=p_b}^{N-p_b} \| -\mathbf{Q}_i + 3\mathbf{Q}_{i+1} - 3\mathbf{Q}_{i+2} + \mathbf{Q}_{i+3}\|^2 \tag{3}$$

where $p_b \geq 3$ is the order of the B-spline, $\mathbf{Q}_i$ represents the position of the ith control point, and N is the total number of control points.

Moreover, the constraints on the waypoints are essential for preserving the original shape of the centerline by employing the waypoint constraint cost f_w, which is defined as the summation of the squared deviations of the waypoints on the current trajectory from the original waypoints W_i as follows:

$$f_w = \sum_{i=p_b}^{N-p_b} \|\text{EvalBspline}(\mathbf{Q}_i, \ldots, \mathbf{Q}_{i+p_b-1}) - W_i\|^2 \tag{4}$$

where the EvalBspline function evaluates the waypoints on the B-spline according to the corresponding control points.

Furthermore, an interval constraint cost f_i is added to prevent fluctuating distances between control points by penalizing deviations in control point speed from a virtual speed v_{virtual} as follows:

$$f_i = \sum_{i=p_b}^{N-1} \left(\| \frac{\mathbf{Q}_{i+1} - \mathbf{Q}_i}{\delta t} \| - v_{\text{virtual}} \right)^2 \tag{5}$$

where δt is the virtual time interval between the control points.

In addition, due to occasional partial observation of the tunnel wall, a distance constraint cost f_d is implemented to enhance safety by quadratically penalizing control points that are closer to the tunnel wall than a predefined distance d_0 as follows:

$$f_d = \sum_{i=p_b}^{N-p_b} F_d(\text{edf}(\mathbf{Q}_i)) \tag{6}$$

where $\text{edf}(\mathbf{Q}_i)$ represents the EDF value at $\mathbf{Q}_i$ and F_d is a differentiable quadratic cost function defined as follows:

$$F_d(\text{edf}(\mathbf{Q}_i)) = \begin{cases} (\text{edf}(\mathbf{Q}_i) - d_0)^2 & \text{edf}(\mathbf{Q}_i) \leq d_0 \\ 0 & \text{edf}(\mathbf{Q}_i) > d_0. \end{cases} \tag{7}$$

Note that d_0 is typically set slight larger than the radius of the quadrotor to ensure a safety margin.

Finally, the end state constraint cost f_e is included to preserve the original terminal states as follows:

$$f_e = \sum_{i=0}^{k} \|\text{EvalBspline}_i(\mathbf{Q}_{N-p_b}, \ldots, \mathbf{Q}_{N-1}) - E_i\|^2 \tag{8}$$

where the EvalBspline_i function computes the ith derivative of the point on the B-spline based on the corresponding control points, and E_i denotes the ith derivative of the original ending state.

A limited-memory BFGS optimization is applied to the control points $\mathbf{Q}_i$'s based on the total cost $f_{@cl}$. This process establishes a smooth tunnel-following trajectory, represented by the optimal control points, for subsequent planning procedures.

2) Tunnel Cross-Section Recognition: Based on the obtained smooth tunnel-following trajectory, the cross-section planes can be determined for recognition, enabling the generation of the corridor. Initially, raw cross-section images are generated along the tunnel-following trajectory according to the local EDF map, as illustrated in Fig. 12(b). Subsequently, these cross-section images are extracted for shape recognition. Two example results are depicted in Fig. 12(c). The recognition pipeline is divided into two stages: shape classification and pattern detection. During this process, we assume that all the shapes of the cross-sections belong to a close set and we recognize the shape patterns within this set. In this article, we focus on rectangles and circles for the shape set, as these cover a wide range of real-world tunnels. Furthermore, the pipeline can be easily extended to arbitrary shapes with minor modifications.

a) Cross-section Shape Classification: In the shape classification stage, we adopt a convolutional neural network (CNN) with two convolutional layers, two pooling layers, and two fully connected layers. The CNN takes the cross-section images as input and outputs the shape class. The input images are generated with resolutions of 35×35 in practice. The training set and validation set are generated from perfect circles and rectangles with random noise and occlusions on the edges. The training set consists of 24 000 images of rectangles and circles in different sizes and orientations at different positions, while the validation set comprises 6000 images of generated rectangles and circles. The test set is formed from 2000 cross-section images collected from real flights. The example images for training, validation and testing are shown in Fig. 13. After training, the accuracy of the classification reaches 99.61% on the validation set and 99.55% on the test set. For extendibility, the CNN can be easily modified to support a larger set of shape classes with more generated images of different shapes.

b) Cross-section Pattern Detection: Once the cross-section shape class is determined by the classification CNN, the detection of the shape pattern can be executed. For rectangles, we

Authorized licensed use limited to: Technion Israel Institute of Technology. Downloaded on September 11,2026 at 02:01:38 UTC from IEEE Xplore. Restrictions apply.

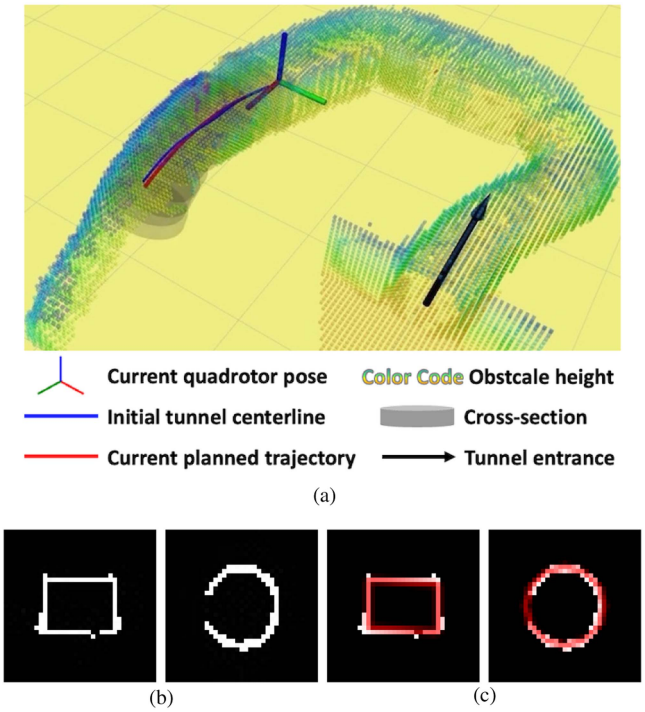


Fig. 12. Visualization and two of the recognized cross-sections when the quadrotor fly about the position where the tunnel shape and size changes in the 2-D tunnel case 3, shown in Fig. 1(c). The detected rectangular and circular cross-sections are marked in red and shown half-transparently in the visualization. (a) Visualization during flight in the tunnel, shown in Fig. 1(c), when cross-sections of different shapes appear in the planning horizon. The color coding indicates the height of the real-time constructed map, and the black arrows show the estimated tunnel entrance. The axes indicate the current pose of the quadrotor. The blue and red curves indicate the initial tunnel centerline from the extracted waypoints and the tunnel-following trajectory, respectively. (b) Raw cross-section images. (c) Detected rectangle and circle.

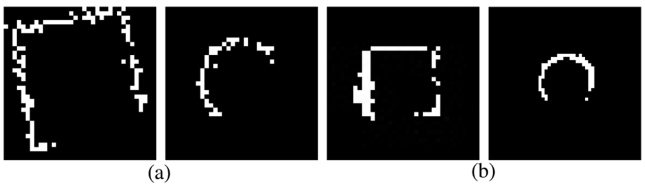


Fig. 13. Example images for training, validation, and testing of the cross-section shape classification CNN. (a) Generated image examples from the training and validation sets. (b) Extracted cross-sections from real flights for testing.

adopt a Hough line-based detection method modified from [38]. After acquiring the parallel line pairs through an enhanced Hough line transform, the rectangle patterns are identified based on right angle and edge length criteria. The final rectangle selection criterion is modified to be a weighted sum of edge intensity, parallelism, and perpendicularity of the edges. For circles, a circle Hough transform is employed for detection. If no pattern of the classified shape is detected, this typically indicates an incomplete map, and the cross-section shape and size are assumed to match the previous one. If a cross-section is of an unknown shape that is neither rectangle nor circle, it is treated as a circle centered at the maximum EDF value point to

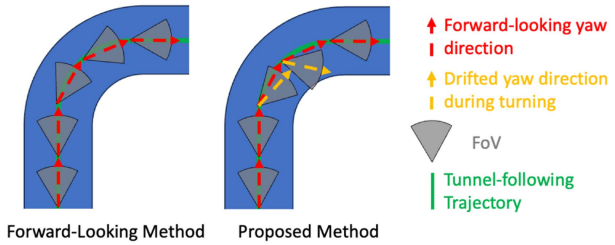


Fig. 14. Illustration of the traditional forward-looking yaw method and the proposed active yaw method. The proposed active yaw method can perceive more information on the inner side at the turn.

maintain sufficient distance from the tunnel wall. The detection process is also easily extendable to more complex shapes using the generalized Hough transform.

3) Tunnel-Following Trajectory Refinement: Following the cross-section recognition, the recognized cross-section centers may deviate from the previous extracted tunnel center waypoints. In such instances, the waypoints are adjusted to match the recognized cross-section centers. Subsequently, the procedures starting from the tunnel-following trajectory optimization can be repeated to acquire a refined tunnel-following trajectory. This ensures that the generated trajectories accurately capture the necessary geometric data of the tunnel while considering the smooth system dynamics.

B. Perception-Aware Active Yaw Planning

With the extracted tunnel-following trajectory, the flight path is determined. However, active yaw planning is still needed for the virtual omnidirectional perception module to sense the environment ahead. This planning involves two stages: yaw waypoint selection and yaw trajectory optimization.

1) Yaw Waypoint Selection: In most general planning frameworks, the yaw direction is set to point toward a position at a constant distance or time ahead on the planned trajectory. This strategy is generally effective in open areas. However, in extremely restricted environments such as narrow tunnels, this approach is inadequate, particularly when the quadrotor needs to perform sharp turns. As illustrated in Fig. 14, the limited FoV often cannot cover the inner side of a turning trajectory. To address this, we employ a side-drifting method for the quadrotor to navigate corners, enhancing visibility of the inner side.

For each sampled waypoint on a given tunnel-following trajectory, a forward-looking 2-D direction $\mathbf{y}_{\mathbf{o}_i}$ can be easily derived, pointing toward a position a constant distance ahead. The drifted yaw direction $\mathbf{y}_{\mathbf{d}_i}$ is then calculated based on the radius $\mathbf{r}_{\mathbf{h}i}$ of the curvature at the sampled waypoint on the 2-D tunnel-following trajectory projected on the horizontal plane and the forward-looking direction $\mathbf{y}_{\mathbf{o}_i}$ as follows:

$$\mathbf{y}_{\mathbf{d}_i} = \mathbf{y}_{\mathbf{o}_i} + k_r \frac{\mathbf{r}_{\mathbf{h}i}}{||\mathbf{r}_{\mathbf{h}i}||^2} \tag{9}$$

where k_r is a constant used to adjust the drift ratio. Utilizing this formulation, the larger the curvature, or the smaller the radius at a

Authorized licensed use limited to: Technion Israel Institute of Technology. Downloaded on September 11,2026 at 02:01:38 UTC from IEEE Xplore. Restrictions apply.

point on the trajectory, the greater the yaw drift at a turn. In addition, when the quadrotor reaches a vertical section of the tunnel, the yaw remains unchanged until the 3-D forward-looking direction deviates sufficiently from the vertical direction. With the 2-D yaw directions $\mathbf{y_d}_i$, the 1-D yaw waypoints wy_i can be simply obtained.

2) Yaw Trajectory Optimization: From the 1-D yaw waypoints wy_i, a B-spline can be parameterized for further optimization. Similar to the approach in Section VI-A1, the total objective is formulated as a weighted sum of the yaw smoothness cost f_{sy}, the yaw waypoint cost f_{wy}, and the yaw end state cost f_{ey} as follows:

$$f_y = \lambda_{\text{sy}} f_{\text{sy}} + \lambda_{\text{wy}} f_{\text{wy}} + \lambda_{\text{ey}} f_{\text{ey}}. \tag{10}$$

The smoothness yaw cost is defined as a third-order elastic band cost as follows:

$$f_{\text{sy}} = \sum_{i=p_b}^{N-p_b} (-qy_i + 3qy_{i+1} - 3qy_{i+2} + 3qy_{i+3})^2 \tag{11}$$

where qy_i represents the ith control point of the parameterized B-spline.

The yaw waypoint cost is similarly defined as follows:

$$f_{\text{wy}} = \sum_{i=p_b}^{N-p_b} k_{wyi} \cdot (\text{EvalBspline}(qy_i, \ldots, qy_{i+p_b-1}) - \text{wy}_i)^2$$
$$k_{wyi} = \begin{cases} 0 & \text{wy}_i \text{ in vertical sections} \\ 1 & \text{otherwise} \end{cases} \tag{12}$$

where the EvalBspline function evaluates the yaw waypoint on the B-spline according to the corresponding control points, except that the deviations from the yaw waypoints in the vertical sections are not penalized.

Finally, the yaw end state cost is defined as follows:

$$f_{\text{ey}} = \sum_{i=0}^{k} (\text{EvalBspline}_i(qy_{N-p_b}, \ldots, qy_{N-1}) - \text{ey}_i)^2 \tag{13}$$

where the EvalBspline_i function evaluates the ith derivative of the point on the B-spline according to the corresponding control points, and ey_i indicates the ith derivative of the original yaw ending state, respectively.

C. Perception-and-Disturbance-Aware Speed Profile Planning

Using the obtained tunnel-following trajectory and the yaw trajectory, the perception-and-disturbance-aware speed profile planning can be executed by leveraging the models formulated in Section V to address both factors. Given the fixed trajectory shape, the planning is conducted within the 1-D distance space of the tunnel-following trajectory to generate speed profiles. This process involves two key phases: initial speed profile search and speed profile optimization.

1) Initial Speed Profile Search: For the initial speed profile search, a hybrid-state A$*$ search [39] along the tunnel-following trajectory is adopted. Multiple discretized acceleration inputs are utilized to generate motion primitives at different time steps according to the state transition equation as follows:

$$\mathbf{x}(t) = \exp(\mathbf{A}t)\mathbf{x}(0) + \int_0^t \exp \mathbf{A}(t-\tau)\mathbf{B}\mathbf{u}(\tau)\, d\tau. \tag{14}$$

The total actual cost $\mathcal{G}$ is defined as a weighted combination of the input, perception, ego airflow disturbance, and time cost as follows:

$$\mathcal{G}(T) = \int_0^T \|\mathbf{u}(t)\|^2 + \lambda_{\text{of}} \hat{v}_{\text{of}}(t)^2 + \lambda_{\text{ed}} \|\text{EDL}(t)\|^2 dt + \rho T \tag{15}$$

where λ_{of}, λ_{ed}, and ρ are the weight constants, $\hat{v}_{\text{of}}(t)$ is the mean optical flow speed, and $\text{EDL}(t)$ is the estimated ego airflow disturbance level at a position on the tunnel-following trajectory at time t. Note that the integral is approximated using the summation of the interpolated $\hat{v}_{\text{of}}(t)$ and $\text{EDL}(t)$ with small time steps during the search process. In addition, since the yaw trajectory is already set, considering the near-hover movement of the quadrotor, the angular velocity of the camera ω_{cam} in (1) only changes with flight speed at a given position on the tunnel-following trajectory. Consequently, $\hat{v}_{\text{of}}(t)$ only depends on position and flight speed, which is the state during the search process.

When using k cameras, the total estimated optical flow speed $\hat{v}_{\text{of}}(t)$ is calculated similarly to a parallel circuit, as the cameras operate in parallel as follows:

$$\hat{v}_{\text{of}}(t) = \frac{1}{\sum_{i=1}^{k} \frac{1}{\hat{v}_{\text{of}_i}(t)}} \tag{16}$$

where $\hat{v}_{\text{of}_i}(t)$ represents the approximated mean optical flow speed of camera i, as determined using the perception factor model elaborated in Section V-A.

The heuristic cost $\mathcal{H}$ is defined as the minimized cost of the input and the time term in $\mathcal{G}$ utilizing Pontryagin's minimum principle [40] as follows:

$$\begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \frac{1}{T^3} \begin{bmatrix} -12 & 6T \\ 6T & -2T^2 \end{bmatrix} \begin{bmatrix} p_g - p_s - v_s T \\ v_g - v_s \end{bmatrix}$$
$$\mathcal{H}(T) = \min_T \left(\rho T + \left(\frac{1}{3}\alpha^2 T^3 + \alpha\beta T^2 + \beta^2 T \right) \right) \tag{17}$$

where p_s, p_g, and v_s and v_g represent the searching and goal positions and velocities, respectively. The total cost used for the hybrid-state A$*$ search, denoted as $\mathcal{F}$, is simply $\mathcal{G} + \mathcal{H}$. For safety considerations, the ending velocity v_g at the end of the tunnel-following trajectory is always set to 0. After executing the hybrid A$*$ search, the states of the initial 1-D waypoints can be retrieved.

2) Speed Profile Optimization: After the initial speed profile search, an optimization similar to the approach in Section VI-A1 is performed to determine the optimum speed profile. The searched 1-D waypoints are first parameterized into a uniform B-spline to facilitate the optimization.

The total cost of the 1-D trajectory f_1 is defined as the weighted sum of the smoothness jerk and acceleration cost f_{sj} and f_{sa}, the perception cost f_p, the ego airflow disturbance cost

Authorized licensed use limited to: Technion Israel Institute of Technology. Downloaded on September 11,2026 at 02:01:38 UTC from IEEE Xplore. Restrictions apply.

f_{ed}, and the ending state cost f_{e1} as follows:

$$f_1 = \lambda_{\text{sj}} f_{\text{sj}} + \lambda_{\text{sa}} f_{\text{sa}} + \lambda_p f_p + \lambda_{\text{ed}} f_{\text{ed}} + \lambda_{e1} f_{e1}. \tag{18}$$

The smoothness jerk and acceleration cost f_{sj} and f_{sa} are included to prevent abrupt maneuvers along the tunnel-following trajectory and are formulated similarly to the elastic band cost used in previous sections as follows:

$$f_{\text{sj}} = \sum_{i=p_b}^{N-p_b} \left(-q_i + 3q_{i+1} - 3q_{i+2} + 3q_{i+3}\right)^2$$

$$f_{\text{sa}} = \sum_{i=p_b}^{N-p_b} \left(q_i - 2q_{i+1} + q_{i+2}\right)^2 \tag{19}$$

where q_i represents the ith control point of the parameterized B-spline.

The perception cost f_p and the ego airflow disturbance cost f_{ed} are defined on the waypoints as follows:

$$f_p = \sum_{i=p_b}^{N-p_b} \hat{v}_{\text{of}}(\text{EvalBspline}(q_i, \ldots, q_{i+p_b-1}))^2 \tag{20}$$

$$f_{\text{ed}} = \sum_{i=p_b}^{N-p_b} \text{EDL}(\text{EvalBspline}(q_i, \ldots, q_{i+p_b-1}))^2 \tag{21}$$

where $\hat{v}_{\text{of}}$ and EDL represent the disturbance level and the calculated optical flow speed at the waypoint, as described in Sections V and VI-C1, and the EvalBspline function evaluates the waypoint and speed on the B-spline according to the corresponding control points.

The ending state cost f_{e1} is also added to constrain the ending states similar to the previous sections as follows:

$$f_{e1} = \sum_{i=0}^{k} (\text{EvalBspline}_i(q_{N-p_b}, \ldots, q_{N-1}) - e1_i)^2 \tag{22}$$

where the EvalBspline_i function evaluates the ith derivative of the point on the B-spline according to the corresponding control points, and $e1_i$'s indicate the ith derivative of the original ending state, which are the original tunnel-following trajectory length and the 0 ending speed in practice.

VII. Experiments and Results

A. Detailed Implementation

We employ three Intel Realsense L515 Lidar cameras as our primary perception sensors, which capture depth and color images at a frequency of 30 Hz each. In addition, a Bosch BMI088 IMU, integrated into the NxtPX4 flight controller, provides data at a rate of 500 Hz to assist in control and state estimation. The comprehensive software system, depicted in Fig. 3, encompasses modules for perception, planning, and high-level control. This software system is implemented in C++ and runs on an Nvidia Orin NX onboard computer, which operates at 2.0 GHz with a maximum power consumption of 25 W. The RGBD-inertial state estimation module outputs 30 Hz camera rate odometries and 500 Hz IMU rate odometries. The mapping module updates at 15 Hz, while the tunnel center waypoint extraction and planning modules both run at 10 Hz. The high-level control module generates thrust and attitude commands at a frequency of 100 Hz, which are then transmitted to the NxtPX4 flight controller to produce the necessary low-level commands executed by the four T-Motor F60 PRO KV2550 motors with 5-in 5045 propellers.

TABLE II
Experiment Sets Configuration

Case	Length	Cross-section shape	Cross-section dimension
Straight	6 m	Square	0.6 m
2-D case 1	10 m	Circle	0.6 m – 0.7 m
2-D case 2	10 m	Circle	0.6 m – 0.7 m
2-D case 3	20 m	Circle and rectangle	0.5 m – 0.7 m
3-D case 1	10 m	Circle	0.6 m – 0.7 m
3-D case 2	23 m	Circle and rectangle	0.5 m – 0.9 m
Rigid vent pipe	20 m	Rectangle	0.5 m – 0.8 m
Construction site	20 m	Circle	0.7 m – 0.8 m

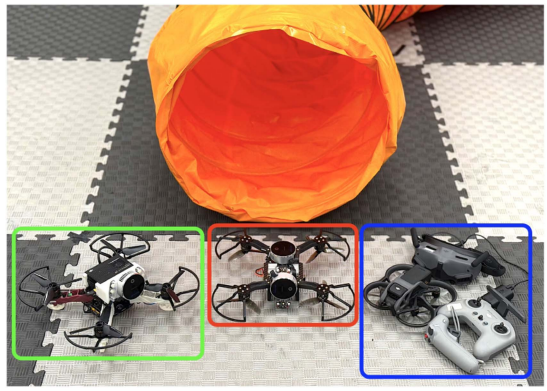


Fig. 15. Autonomous tunnel flight system (red frame), the tunnel flight system in [6] (green frame), and the commercial DJI Avata FPV system for manual flights (blue frame).

B. Experiment Sets Configuration

Multiple sets of experiments are conducted in various scenarios (see Fig. 1), covering both 2-D and 3-D cases, with details provided in Table II. We use flexible and rigid pipes made from plastic and steel, featuring various cross-sectional shapes and dimensions to represent a wide range of real-world situations.

In addition, we conducted a series of ablation studies on the proposed planning method in selected representative cases. We also performed comparisons with the system in [6] and manual flights performed by an experienced pilot, in both 2-D and 3-D tunnels. For manual flights, we use the DJI Avata, which is one of the most advanced and compact commercial FPV systems, featuring a 25 cm diameter. The system in [6], the commercial DJI Avata FPV system, and the proposed system are illustrated in Fig. 15.

C. Straight Tunnel

The first two sets of experiments are conducted in a straight 6-m-long tunnel with a squared cross-sectional shape of 0.6 m edge length, as shown in Fig. 16, to examine the characteristics of the proposed system.

Authorized licensed use limited to: Technion Israel Institute of Technology. Downloaded on September 11,2026 at 02:01:38 UTC from IEEE Xplore. Restrictions apply.

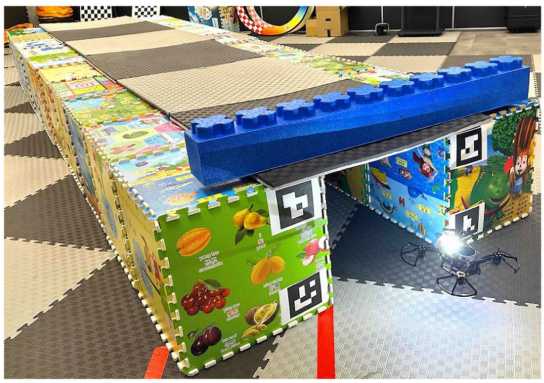


Fig. 16. Straight tunnel with a squared cross-sectional shape of 0.6 m edge length for flight tests.

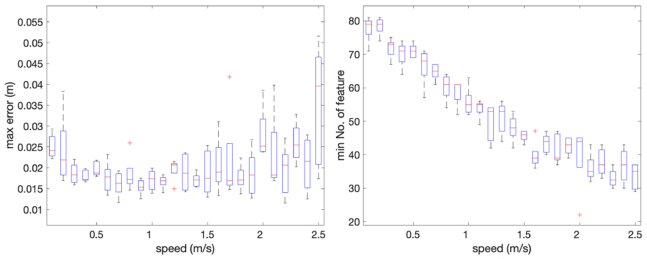


Fig. 17. Box plots of the maximum control tracking error on the cross-section planes and the minimum number of features that can be tracked by the state estimation system during the flights in the straight tunnel shown in Fig. 16. (a) Maximum control tracking error. (b) Minimum number of features.

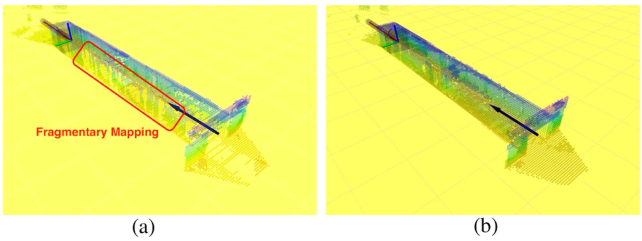


Fig. 18. Real-time constructed map of the quadrotor flying through the straight tunnel shown in Fig. 16. Fragmentary mapping results during the slow speed flight can be easily recognized in the red rectangle. (a) Slow flight speed at 0.2 m/s. (b) Proposed method.

The first set of experiments consist of 125 flights are performed at constant flight speeds ranging from 0.1 m/s to 2.5 m/s in 0.1 m/s intervals, with each speed tests five times on the proposed quadrotor platform. As depicted in the control tracking error and feature number plots (see Fig. 17), the results align with previous studies [6] despite configuration differences. Large tracking errors occur at slow speeds due to ego airflow disturbances and at high speeds due to modeling errors and actuator limitations. In addition, the minimum number of tracked features decreases as speed increases due to large parallax and motion blur. Furthermore, the experiments highlight that slow speeds can cause shaking movements, leading to fragmentary mapping results, as shown in Fig. 18(a), which are unfavorable during tunnel flights. The consensual results underscore the necessity of

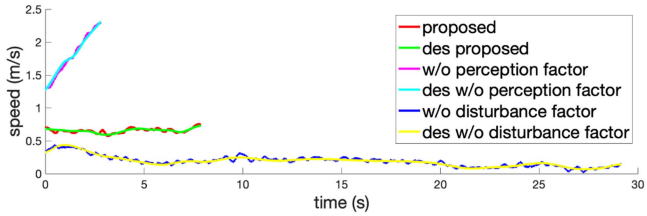


Fig. 19. Speed profiles in the ablation study during straight tunnel flights.

compensating for perception and ego airflow disturbance factors, validating the motivations behind this work.

The second set of experiments involve ablation studies on the perception and ego airflow disturbance costs in the same straight tunnel. The proposed method and the ablation of either cost are each tested five times to gather control tracking error and feature tracking results. As shown in Table III, the proposed method achieves the minimum cross-section tracking error compared to both ablation scenarios. Note that we mainly focus on the errors in cross-sections since small longitudinal errors generally do not cause safety issues. Notably, the results also indicate that the proposed method attains comparable feature tracking numbers with the ablation of the disturbance cost and significantly higher numbers compared to the ablation of the perception cost. This is attributed to the automatic compensation of the perception and disturbance factors, ensuring that the quadrotor flies neither too fast or too slow through the tunnel, as depicted in Fig. 19. The appropriate flight speed leads to a moderate optical flow speed and trajectory duration. These results preliminarily verify the functionality of the system in straight tunnels.

D. 2-D Tunnels

The next three sets of experiments are conducted in three different 2-D tunnels constructed using one or more flexible vent pipes commonly employed in industrial settings, as shown in Fig. 1(a)–(c).

We first perform an ablation study focusing on ego airflow disturbance, perception costs, and perception-aware active yaw planning in the 2-D tunnel case 1. This scenario, shown in Fig. 1(a), features a flexible vent pipe with circular cross-sections ranging from 0.6 to 0.7 m in diameter. Each configuration is tested five times to avoid accidental results, and an overview on the visualization result using the proposed method is shown in Fig. 20(a).

According to the results in Table III, removing the perception cost consistently leads to failures at the first turn. This is because the increased ego airflow disturbance causes the quadrotor to accelerate until it completely avoids the ego airflow, regardless of the tunnel's curvature. Conversely, removing the ego airflow disturbance cost leads to unnecessarily slow flight speeds within the tunnel, causing the quadrotor to shake due to the turbulent flow it generates. The proposed perception-and-disturbance-aware speed profile planning automatically compensates for the two factors by reducing speed at turns and appropriately increasing speed in sections with small curvatures [see Fig. 20(b)]. This

Authorized licensed use limited to: Technion Israel Institute of Technology. Downloaded on September 11,2026 at 02:01:38 UTC from IEEE Xplore. Restrictions apply.

TABLE III
ABLATION STUDY ON PERCEPTION AND EGO AIRFLOW DISTURBANCE COSTS

Seq.	Straight			2-D tunnel case 1			3-D tunnel case 1		
	Proposed	$\lambda_p = 0$	$\lambda_{\text{ed}} = 0$	Proposed	$\lambda_p = 0$	$\lambda_{\text{ed}} = 0$	Proposed	$\lambda_p = 0$	$\lambda_{\text{ed}} = 0$
Max error in cross-section (m)	**0.0203**	0.0301	0.0228	**0.0401**	×	0.0407	**0.0537**	×	0.0554
Min No. feature	72.8	43.1	**78.4**	**32.2**	×	29.4	36.2	×	**53.2**
Avg. optical flow speed (pixel/s)	0.3935	0.8085	**0.2811**	0.5071	×	**0.423**	0.1858*	×	**0.1497***
Traj. duration (s)	13.411	**5.9076**	32.7925	**24.103**	×	52.6244	**20.2722**	×	31.207

×: fail. **Bold**: best results. *:Using three cameras and calculated by 16.

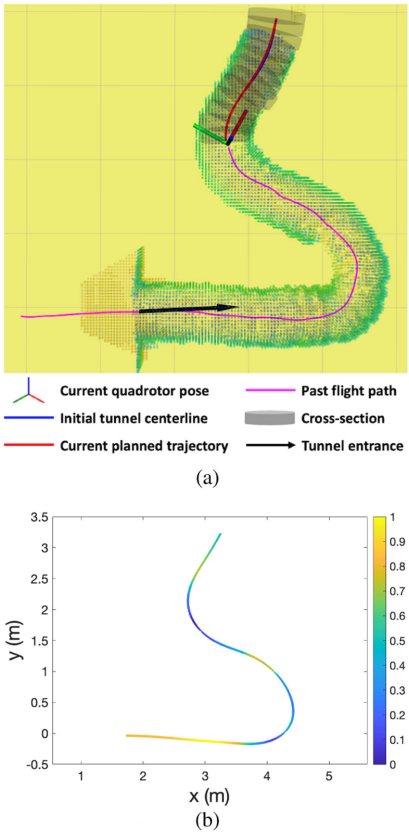


Fig. 20. Visualization screenshots and the speed profile of the quadrotor flying through the 2-D tunnel case 1 shown in Fig. 1(a). The color coding in the visualization indicates the height of the real-time constructed map, and the black arrow shows the estimated tunnel entrance; the axes indicate the current pose of the quadrotor, while the pink, blue, and red lines indicate the past flight path, the initial tunnel centerline, and the current planned tunnel-following trajectory, respectively. The extracted and predicted cross-sections are shown half-transparently. (a) Visualization. (b) Speed profile.

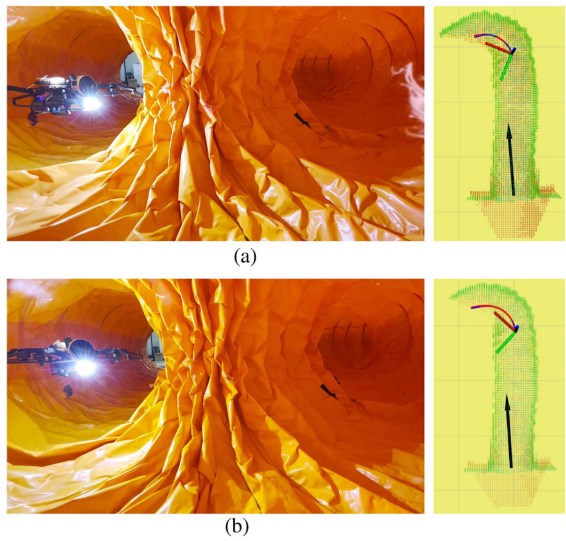


Fig. 21. Snapshots of the quadrotor flying through the first corner using the proposed method and the method without active yaw planning in the 2-D tunnel case 1 shown in Fig. 1(a). The markers are the same as previous figures. It can be clearly seen that the quadrotor using the proposed method turns the yaw at a larger angle. (a) Proposed method. (b) Without active yaw planning.

approach achieves a minimum control tracking error and maintains a sufficient number of tracked features for state estimation. In addition, it achieves higher efficiency in terms of traversal time compared to configurations without ego airflow disturbance cost. These results demonstrate the effectiveness of the proposed speed profile planning. Furthermore, we compared the proposed method with the ablation of perception-aware active yaw planning, where the yaw direction always points toward a point at a constant distance ahead on the trajectory. In five trails without the active yaw planning, the quadrotor consistently stops at the first turn because the yaw headings never turn enough to provide an adequate view of the tunnel ahead. In contrast, the proposed method leverages active yaw planning to successfully traverse the tunnel in all five trials. The heading differences during one of the comparison experiments are clearly illustrated in Fig. 21.

We then compare the proposed system with the system in [6] and manual flight using a commercial FPV system, as shown in Fig. 15, in another 2-D tunnel using the same vent pipe yet of a different shape, as shown in Fig. 1(b). Each configuration is tested five times. The snapshots and visualization results of the proposed method are shown in Fig. 22.

During the five flights using the proposed system, no failures are reported, while the system in [6] always crashes at the first shape turn despite the known 0.7 m tunnel diameter and a proper flight speed of 1 m/s from its previous results. The system in [6] is unable to handle the large curvature and narrower dimensions caused by the folds of the pipe at the turn in all five trails, as it lacks cross-section recognition, active yaw planning, and speed profile planning. In addition, the FPV drone operated by an experienced pilot is kept sucking to the tunnel wall during the manual flight and eventually crashes inside the tunnel in all five trails, despite its smaller size compared to the proposed system. This set of experiment demonstrates the superior performance of the proposed system against the system in [6] and experienced human pilots.

Authorized licensed use limited to: Technion Israel Institute of Technology. Downloaded on September 11,2026 at 02:01:38 UTC from IEEE Xplore. Restrictions apply.

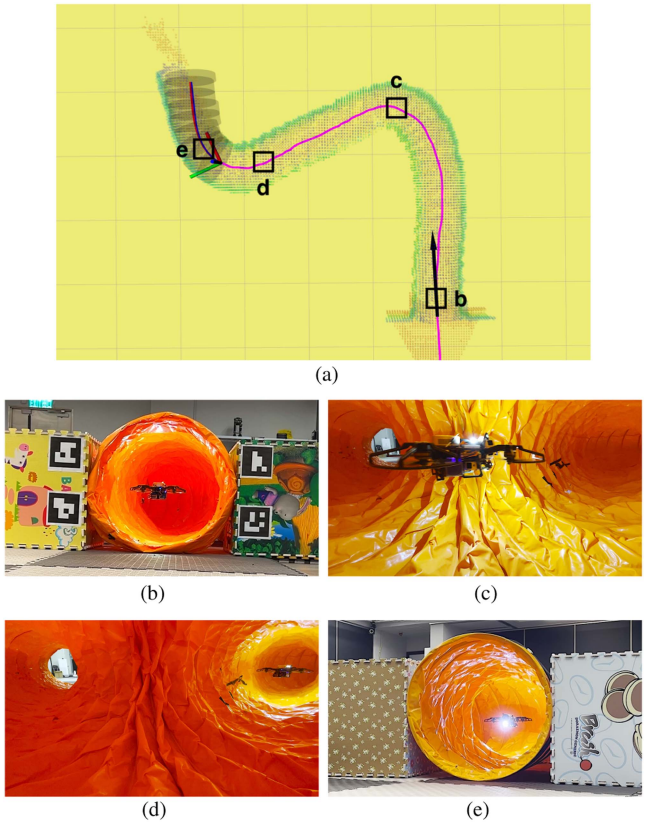


Fig. 22. Snapshots and visualization results when the quadrotor flying through the 2-D tunnel shown in Fig. 1(b). The markers are the same as previous figures and the quadrotor positions in the snapshots are labeled in the visualization. (a) Visualization. (b) Flying into the entrance. (c) Flying through turn 1. (d) Flying through turn 2. (e) Flying out of the exit.

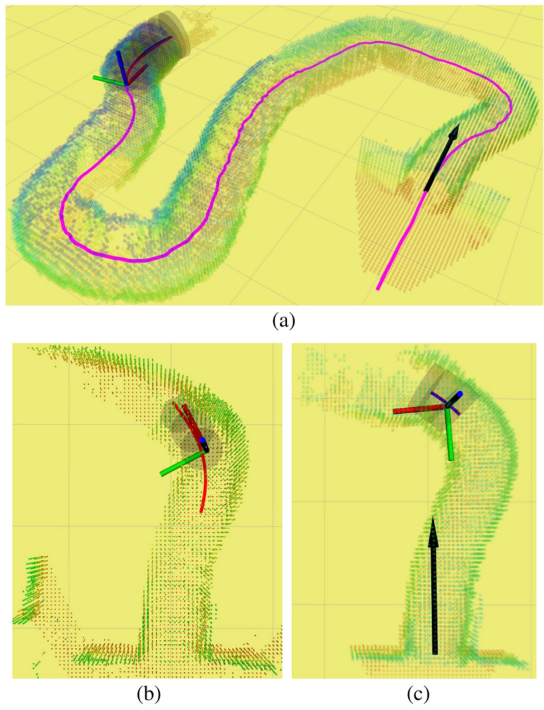


Fig. 23. Visualization screenshots of the quadrotors flying through the 2-D tunnel shown in Fig. 1(c). The markers are same as previous figures. The system in [6] crashes at the first turn and the constant speed method emergency stops at the first turn. (a) Overview from the proposed system. (b) [6]. (c) Proposed system with constant speed of 0.7 m/s.

The final set of 2-D tunnel experiments is conducted using a series of flexible vent pipes with varying cross-sectional shapes: a circular section with diameter of 0.5 to 0.55 m, a rectangular section measuring 0.5 m $\times$ 0.65 m, and another circular section with diameter of 0.6 to 0.7 m, as illustrated in Fig. 1(c). The performance of the proposed system is evaluated against the system described in [6], manual flights by an experienced pilot, and the ablation of ego airflow disturbance and perception-aware speed profile planning, also known as the constant speed method. Each configuration is tested five times, except that more trails are conducted on manual flights.

As shown in Fig. 23(a), the proposed system successfully navigates through different sections and sharp turns in all five trials, demonstrating its robustness and adaptability. In contrast, as shown in Fig. 23(b), the system in [6] fails at the first turn due to its inability to manage changes in cross-sectional dimensions and sharp turns, even at a flight speed of 1 m/s. Furthermore, despite the pilot's experience from previous tunnels, the manual flight crashes in all ten trials before the first turn of the tunnel, regardless of using a joystick motion controller or a conventional radio controller. This failure is attributed to stronger ego-airflow disturbances caused by the smaller cross-section size of the tunnel. The quadrotor is immediately sucked to the tunnel wall even when it is just several centimeters away from the centerline and there is almost no chance to recover. Although the constant speed method using the proposed system at 0.7 m/s does not result in crashes, it requires five emergency stops at the first turn due to safety concerns. The flight speed at the sharp turn is excessively fast, preventing timely map updates. These results highlight the proposed system's capability to handle tunnels with varying cross-sections and sharp turns effectively. Comparisons with the system in [6], manual flight and the constant speed method further underscore the necessity and effectiveness of cross-section recognition, active yaw, and speed profile planning.

E. 3-D Tunnels

Using the proposed system, we conduct further two sets of experiments in 3-D narrow tunnels with level variations, including vertical sections, as illustrated in Fig. 1(d)–(e).

The first set of flights involves an ablation study on the perception and ego airflow disturbance costs within a flexible vent pipe of 0.6 to 0.7 m diameter, previously used and reshaped to include two vertical sections, as shown in Fig. 1(d). Each configuration is tested five times, similar to prior experiments, and a visualization screenshot is provided in Fig. 24. As indicated in Table III, removing the perception cost results in failures in all five trials due to excessive speed at the first turn. Conversely, the ablation of the ego airflow disturbance cost leads to unnecessary slow flight speeds, thereby reducing efficiency in terms of traversal time. The proposed method automatically adjusts the speed by

Authorized licensed use limited to: Technion Israel Institute of Technology. Downloaded on September 11,2026 at 02:01:38 UTC from IEEE Xplore. Restrictions apply.

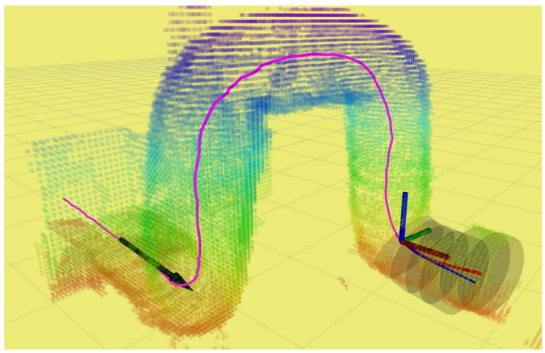


Fig. 24. Screenshot of the visualization during flight of the proposed system inside 3-D tunnel case 1 shown in Fig. 1(d). The markers are same as previous figures.

balancing both factors, achieving optimal control performance and the shortest flight time while maintaining sufficient tracked features for state estimation. This results in smooth and efficient traversal in all five trials. This set of ablation studies demonstrates the effectiveness of the proposed perception and ego airflow disturbance models and the entire system in tunnels with level variations, including vertical sections.

The second set of flights involves a comparison with manual flight in a more complex 3-D tunnel, as depicted in Fig. 1(e). This tunnel consists of sections with varying slopes and cross-sectional shapes, including circles and rectangles with dimensions ranging from 0.5 m to 0.9 m. The proposed quadrotor system successfully traverses the tunnel in 38 s, with snapshots and visualization results shown in Fig. 25. Conversely, the experienced pilot operating the smaller commercial FPV drone, as shown in Fig. 15, finds it extremely challenging to navigate the same tunnel, with the drone frequently being pulled toward the tunnel walls. The pilot takes 55 s to reach the downslope, as shown in Fig. 25(d), covering only about half of the tunnel length from the entrance, and eventually crashes at that slope. These comparisons demonstrate the superior robustness and adaptivity of the proposed system in complex scenarios, as well as its efficiency, outperforming even experienced pilots.

F. Rigid Vent Pipe

Previous experiments are predominantly conducted in flexible vent pipes. To validate the generality of the system, we also conduct experiments in a rigid vent pipe with rectangular cross-sections, ranging from 0.5 m to 0.8 m, and featuring right angle sharp turns, as shown in Fig. 1(f). The proposed system takes 66 s to successively traverse this narrow vent pipe. The snapshots and visualization results during the flight are shown in Fig. 26. To the best of the authors' knowledge, no previous work has demonstrated the capability to navigate such a narrow vent pipe. This is the first system capable of being adopted in such a 3-D narrow tunnel. This experiment showcases the system's exceptional robustness and capability.

G. Vent Pipe on a Construction Site

To further validate the system in a real-world setting, we also conduct flight experiments in a vent pipe on an active construction site, transitioning from an indoor area to an outdoor area, as shown in Fig. 1(g). The vent pipe is 20 m long, with a circular cross-section ranging from 0.7 to 0.8 m in diameter. The visualization result and snapshots during flights are shown in Fig. 27. Despite the strong dust on the site and in the pipe that can be clearly seen in Fig. 27(c) and (b), the proposed autonomous quadrotor system is still able to traverse the pipe smoothly from indoor to outdoor. This experiment on the construction site demonstrates the adaptivity and robustness of the system and its potential for industrial applications in the future.

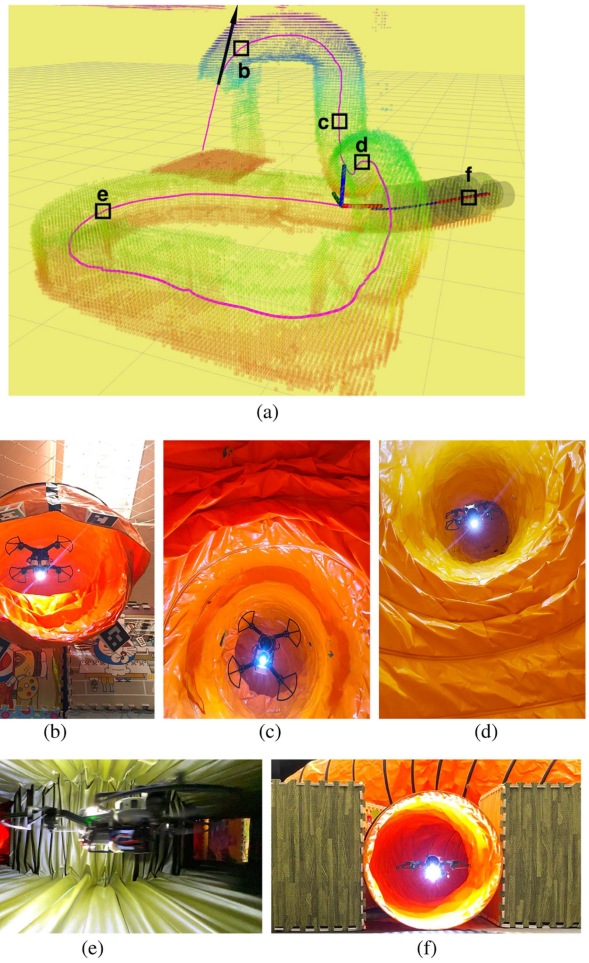


Fig. 25. Snapshots and visualization results when the quadrotor flies through the 3-D tunnel case 2 shown in Fig. 1(e). The markers are same as previous figures and the quadrotor positions in the snapshots are labeled in the visualization. (a) Visualization. (b) Flying into the entrance. (c) Flying down the vertical section. (d) Flying upward along the slope. (e) Traversing the dark rectangular tunnel section. (f) Flying out of the exit.

H. Summary on Traversalibilities of Different Systems

As indicated in [6], state-of-the-art generalized motion planners not designed for navigating narrow tunnels, for instance [7], are not able to navigate the quadrotor through a narrow tunnel. Therefore, comparisons with them are not meaningful. Moreover, it is observed that during manual flights, the quadrotor tends to veer toward the tunnel walls and is difficult to recover, even when just centimeters away from the centerline. This suggests that planners not following the centerline of narrow

Authorized licensed use limited to: Technion Israel Institute of Technology. Downloaded on September 11,2026 at 02:01:38 UTC from IEEE Xplore. Restrictions apply.

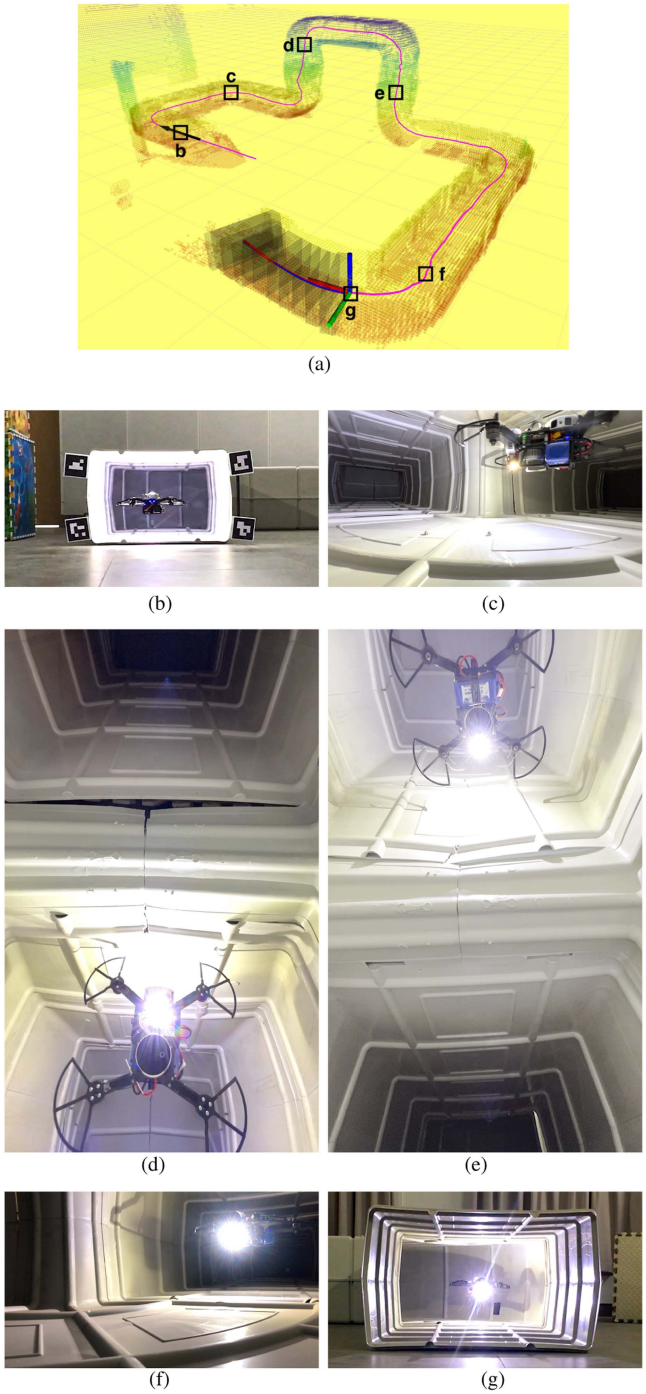


Fig. 26. Snapshots and visualization results when the quadrotor flies through a rigid vent pipe shown in Fig. 1(f). The markers are same as previous figures and the quadrotor positions in the snapshots are labeled in the visualization. (a) Visualization. (b) Flying into the entrance. (c) Flying through turn 2. (d) Flying up the first vertical section. (e) Flying down the second vertical section. (f) Flying through the last turn. (g) Flying out of the exit.

tunnels [29] or systems with large control tracking errors [30] are not capable of navigating narrow tunnels effectively. Therefore, the comparisons with those planners or systems also hold little significance.

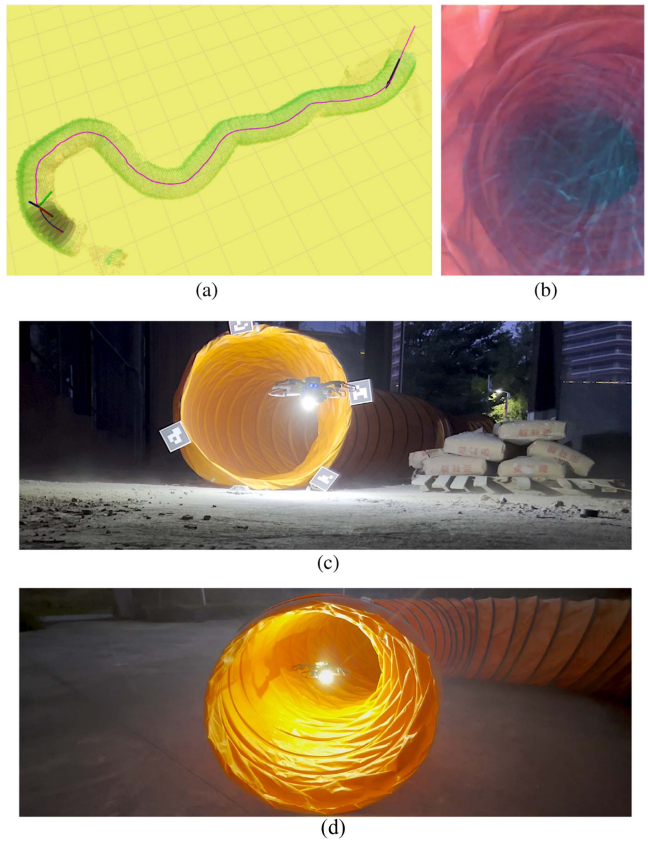


Fig. 27. Snapshots and visualization results when the quadrotor flies through a vent pipe on a construction site shown in Fig. 1(g). The markers are same as previous figures and the quadrotor positions in the snapshots are labeled in the visualization. (a) Visualization. (b) Onboard camera view during the flight. (c) Flying into the entrance. (d) Flying out of the exit.

TABLE IV
COMPARISON ON TRAVERSALIBILITIES OF DIFFERENT SYSTEMS IN 2-D AND 3-D TUNNELS

System	2-D tunnel case 2	2-D tunnel case 3	3-D tunnel case 2
Proposed	○	○	○
[6]	×	×	–
Manual	×	×	×

○: success in all trails. ×: fail in all trails. –: not supported.

As a result, we compare the proposed system with others that are intuitively capable of navigating narrow tunnels in more general scenarios. In summary, we compared the proposed system with the system in [6] and manual flights by an experienced pilot in 2-D tunnel case 2, case 3, and 3-D tunnel case 2. Five flights are conducted for each configuration, except that additional manual flights are performed in certain cases. As shown in the traversal results in Table IV, the proposed system successfully navigates all scenarios, whereas the system in [6] and manual flights fail to traverse any of the tunnels. The system in [6] always fails in 2-D tunnel cases 2 and 3 due to its inability to manage sharp turns and variations in cross-sectional shapes and sizes. In addition, it lacks support for 3-D tunnels because of inadequate perception capabilities in upward and downward directions. Manual flights also struggle with these tunnels, no matter the pilot is using a

Authorized licensed use limited to: Technion Israel Institute of Technology. Downloaded on September 11,2026 at 02:01:38 UTC from IEEE Xplore. Restrictions apply.

conventional controller or a motion controller, as maintaining the drone along the tunnel centerline is extremely challenging. Any slight deviation results in the drone being sucked into the tunnel wall, leaving almost no chance for recovery in such narrow spaces. For the 2-D tunnel case 3, manual flight cannot even make to the first turn in all ten trails. These outcomes highlight the superior performance and capability of the proposed system over the system in [6] and experienced human pilots.

VIII. Extendability of the System and Discussion

In the experiments and results, the proposed autonomous aerial system demonstrates exceptional robustness in navigating a wide range of narrow tunnels with both circular and rectangular cross-sections. The proposed system can be extended to other tunnel-like environments with various cross-sectional shapes. In addition, both the complete system framework and individual modules, including perception, planning, and control, can be transferred to multirotor platforms of different sizes and purposes.

We can also outline the following pipeline to adapt the proposed method for different multirotors and expand its application to tunnels with various cross-sectional shapes:

1) Construct CFD analysis cases according to the multirotor configuration and specification, as well as the possible cross-section shapes of the tunnel to collect the raw ego airflow disturbance data. Then, retrain the MLP disturbance factor model based on the GRNN results derived from the raw data;
2) Configure the perception factor model according to the configuration and parameters of the cameras on the multirotor;
3) Generate images of all possible shapes of the tunnel cross-sections and retrain the classification CNN based on these images. Use the generalized Hough transform to perform cross-section pattern detection for tunnels of general shapes;
4) Deploy the proposed planning module, which includes the updated cross-section shape classification and detection, along with the revised ego airflow disturbance and perception factor models. Then, integrate these with the perception and control modules on the multirotor.

In the current setup, we successfully use a quadrotor with a 40 cm diameter to navigate tunnels with 50 cm diameter cross-sections, achieving an cross-section to quadrotor diameter ratio of 1.25. It is also foreseeable that a quadrotor is able to navigate a tunnel with larger cross-section to quadrotor diameter ratio and lesser ego airflow disturbance. While the diameter ratio is evident for a specific quadrotor and tunnel, analyzing disturbances requires either CFD simulations or real-world tests, which are complex. Cases not meeting these criteria also need further analysis. Developing explicit traversability criteria could be explored in future work.

In addition, the corner shape can also affect the traversability. In the current setting, the quadrotor is able to navigate tunnels with right angles, even U-turns with radius as small as 0.3 m. Except for the physical size constraints, the major constraint is on the FoV of the perception. While the current module with yaw movements and a simple 3-camera system achieves omnidirectional capability, a true omnidirectional perception system offers a stronger FoV. This system eliminates the need for yaw movement during flights, resulting in reduced optical flow speed and potentially enhanced perception performance. It is worth mentioning that a physical omnidirectional setup may increase the payload for the multirotor system, posing significant challenges for hardware design and requiring more computational resources, which could compromise traversability and flexibility.

IX. Conclusion

In this article, we proposed a comprehensive autonomous aerial system designed for navigating narrow tunnels. Extensive flight experiments in various realistic and challenging narrow tunnels validated the robustness of the entire system. Furthermore, a pipeline for deploying the system on other multirotor platforms is provided, facilitating future developments. The proposed system is the first autonomous aerial system capable of smoothly navigating tunnels in arbitrary directions, with sharp corners, and as narrow as 0.5 m in diameter, outperforming experienced human pilots. This work significantly extends the capabilities of MAVs and offers guidelines for future aerial applications in narrow tunnels.

Despite the superior performance of the system, there are still potential improvements in the future as follows:

1) Currently, the system is limited to narrow tunnels without obstacles. We plan to enhance the planning framework to handle obstacles inside tunnels, thereby improving safety and versatility, in future work;
2) The current system can only navigate tunnels without forks. A more advanced planning framework will be introduced in the future to deal with forks in tunnels. This enhancement can also incorporate exploration planning algorithms, which can further improve the traversal and mapping efficiency;
3) The proposed system comprises a single MAV. There is potential to scale this system by incorporating multiple MAVs and considering the interactions between individual drones in terms of perception and airflow disturbances, thereby enhancing the system's efficiency.

These enhancements are aimed at broadening the application scope of the system, including uses in geological prospecting, industrial pipe inspection, search and rescue operations in mine tunnels, etc. These future developments will leverage the capabilities of MAVs to operate in environments that are typically challenging or hazardous for humans.

References

[1] K. Máthé, L. Buşoniu, L. Barabás, C.-I. Iuga, L. Miclea, and J. Braband, "Vision-based control of a quadrotor for an object inspection scenario," in *Proc. IEEE Int. Conf. Unmanned Aircr. Syst.*, 2016, pp. 849–857.
[2] L. Wang et al., "A collaborative aerial-ground robotic system for fast exploration," in *Proc. Int. Symp. Exp. Robot.*, 2018, pp. 59–71.

Authorized licensed use limited to: Technion Israel Institute of Technology. Downloaded on September 11,2026 at 02:01:38 UTC from IEEE Xplore. Restrictions apply.

[3] S. G. Manyam, S. Rasmussen, D. W. Casbeer, K. Kalyanam, and S. Manickam, "Multi-UAV routing for persistent intelligence surveillance reconnaissance missions," in *Proc. Int. Conf. Unmanned Aircr. Syst.*, 2017, pp. 573–580.
[4] T. Özaslan et al., "Autonomous navigation and mapping for inspection of penstocks and tunnels with MAVs," *IEEE Robot. Autom. Lett.*, vol. 2, no. 3, pp. 1740–1747, Jul. 2017.
[5] C. H. Vong, K. Ryan, and H. Chung, "Integral backstepping position control for quadrotors in tunnel-like confined environments," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2019, pp. 6425–6431.
[6] L. Wang, H. Xu, Y. Zhang, and S. Shen, "Neither fast nor slow: How to fly through narrow tunnels," *IEEE Robot. Autom. Lett.*, vol. 7, no. 2, pp. 5489–5496, Apr. 2022.
[7] B. Zhou, F. Gao, L. Wang, C. Liu, and S. Shen, "Robust and efficient quadrotor trajectory generation for fast autonomous flight," *IEEE Robot. Autom. Lett.*, vol. 4, no. 4, pp. 3529–3536, Oct. 2019.
[8] A. Betz, "The ground effect on lifting propellers," *Nat. Advisory Committee Aeronaut.*, Washington, DC, USA, Rep. no. NACA-TM-836, 1937.
[9] I. Cheeseman and W. Bennett, "The effect of ground on a helicopter rotor in forward flight," Ministry Supply, U.K., Aeronautical Res. Council Rep. Memoranda No. 3021, Sep. 1955.
[10] G. M. Eberhart, "Modeling of ground effect benefits for multi-rotor small unmanned aerial systems at hover," *Master's thesis, Dep. Mech. Eng.*, Ohio Univ., Athens, OH, USA, 2017. [Online]. Available: http://rave.ohiolink.edu/etdc/view?acc_num=ohiou1502802483367365
[11] S. A. Conyers, "Empirical evaluation of ground, ceiling, and wall effect for small-scale rotorcraft," Ph.D. dissertation, Dep. Elect. Comp. Eng., Univ. Denver, Denver, CO, USA, 2019. [Online]. Available: https://digitalcommons.du.edu/etd/1570
[12] X. Kan, J. Thomas, H. Teng, H. G. Tanner, V. Kumar, and K. Karydis, "Analysis of ground effect for small-scale UAVs in forward flight," *IEEE Robot. Autom. Lett.*, vol. 4, no. 4, pp. 3860–3867, Oct. 2019.
[13] V. Britcher and S. Bergbreiter, "Use of a MEMs differential pressure sensor to detect ground, ceiling, and walls on small quadrotors," *IEEE Robot. Autom. Lett.*, vol. 6, no. 3, pp. 4568–4575, Jul. 2021.
[14] C. Powers, D. Mellinger, A. Kushleyev, B. Kothmann, and V. Kumar, "Influence of aerodynamics and proximity effects in quadrotor flight," in *Proc. Int. Symp. Exp. Robot.*, Springer, 2013, pp. 289–302.
[15] L. Danjun, Z. Yan, S. Zongying, and L. Geng, "Autonomous landing of quadrotor based on ground effect modelling," in *Proc. 34th Chin. Control Conf.*, 2015, pp. 5647–5652.
[16] W. Johnson, *Helicopter theory*. North Chelmsford, MA, USA: Courier Corporation, 2012.
[17] T. Nishio et al., "Stable control in climbing and descending flight under upper walls using ceiling effect model based on aerodynamics," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2020, pp. 172–178.
[18] D. C. Robinson, H. Chung, and K. Ryan, "Computational investigation of micro rotorcraft near-wall hovering aerodynamics," in *Proc. IEEE Int. Conf. Unmanned Aircr. Syst.*, 2014, pp. 1055–1063.
[19] L. Wang, B. Zhou, C. Liu, and S. Shen, "Estimation and adaption of indoor ego airflow disturbance with application to quadrotor trajectory planning," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2021, pp. 384–390.
[20] E. Mueggler, B. Huber, and D. Scaramuzza, "Event-based, 6-DoF pose tracking for high-speed maneuvers," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2014, pp. 2761–2768.
[21] S. Shen, Y. Mulgaonkar, N. Michael, and V. Kumar, "Vision-based state estimation and trajectory control towards high-speed flight with a quadrotor," in *Proc. Robot. Sci. Syst.*, vol. 1, 2013, p. 1–8.
[22] B. Morrell et al., "Comparison of trajectory optimization algorithms for high-speed quadrotor flight near obstacles," *IEEE Robot. Autom. Lett.*, vol. 3, no. 4, pp. 4399–4406, Oct. 2018.
[23] D. Fridovich-Keil, S. L. Herbert, J. F. Fisac, S. Deglurkar, and C. J. Tomlin, "Planning, fast and slow: A framework for adaptive real-time safe trajectory planning," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2018, pp. 387–394.
[24] L. Quan, Z. Zhang, X. Zhong, C. Xu, and F. Gao, "Eva-planner: Environmental adaptive quadrotor planning," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2021, pp. 398–404.
[25] J. Chen, T. Liu, and S. Shen, "Online generation of collision-free trajectories for quadrotor flight in unknown cluttered environments," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2016, pp. 1476–1483.
[26] G. Loianno, C. Brunner, G. McGrath, and V. Kumar, "Estimation, control, and planning for aggressive flight with a small quadrotor with a single camera and IMU," *IEEE Robot. Autom. Lett.*, vol. 2, no. 2, pp. 404–411, Apr. 2017.
[27] F. Gao, L. Wang, B. Zhou, X. Zhou, J. Pan, and S. Shen, "Teach-repeat-replan: A complete and robust system for aggressive flight in complex environments," *IEEE Trans. Robot.*, vol. 36, no. 5, pp. 1526–1545, Oct. 2020.
[28] M. Petrlík, T. Báča, D. Heřt, M. Vrba, T. Krajník, and M. Saska, "A robust UAV system for operations in a constrained environment," *IEEE Robot. Autom. Lett.*, vol. 5, no. 2, pp. 2169–2176, Apr. 2020.
[29] J. Arrizabalaga and M. Ryll, "Towards time-optimal tunnel-following for quadrotors," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2022, pp. 4044–4050.
[30] T. Elmokadem and A. V. Savkin, "A method for autonomous collision-free navigation of a quadrotor UAV in unknown tunnel-like environments," *Robotica*, vol. 40, no. 4, pp. 835–861, 2022, doi: 10.1017/S0263574721000849.
[31] T. Qin, P. Li, and S. Shen, "VINS-mono: A robust and versatile monocular visual-inertial state estimator," *IEEE Trans. Robot.*, vol. 34, no. 4, pp. 1004–1020, Aug. 2018.
[32] T. Qin, J. Pan, S. Cao, and S. Shen, "A general optimization-based framework for local odometry estimation with multiple sensors," 2019, *arXiv:1901.03638*.
[33] L. Wang, Y. Xu, and S. Shen, "VINS-multi: A robust asynchronous multi-camera-IMU state estimator," 2024, *arXiv:2405.14539*.
[34] L. Han, F. Gao, B. Zhou, and S. Shen, "Fiesta: Fast incremental Euclidean distance fields for online motion planning of aerial robots," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2019, pp. 4423–4430.
[35] D. Specht, "A general regression neural network," *IEEE Trans. Neural Netw.*, vol. 2, no. 6, pp. 568–576, Nov. 1991.
[36] S. Quinlan and O. Khatib, "Elastic bands: Connecting path planning and control," in *Proc. IEEE Int. Conf. Robot. Autom.*, 1993, pp. 802–807.
[37] Z. Zhu, E. Schmerling, and M. Pavone, "A convex optimization approach to smooth trajectories for motion planning with car-like robots," in *Proc. IEEE Conf. Decis. Control*, 2015, pp. 835–842.
[38] C. R. Jung and R. Schramm, "Rectangle detection based on a windowed Hough transform," in *Proc. IEEE 17th Braz. Symp. Comput. Graph. Image Process.*, 2004, pp. 113–120.
[39] D. Dolgov, S. Thrun, M. Montemerlo, and J. Diebel, "Path planning for autonomous vehicles in unknown semi-structured environments," *Int. J. Robot. Res.*, vol. 29, no. 5, pp. 485–501, 2010.
[40] M. W. Mueller, M. Hehn, and R. D'Andrea, "A computationally efficient motion primitive for quadrocopter trajectory generation," *IEEE Trans. Robot.*, vol. 31, no. 6, pp. 1294–1310, Dec. 2015.



Luqi Wang received the B.Eng. degree in computer engineering and aerospace engineering and the Ph.D. degree in electronic and computer engineering from the Hong Kong University of Science and Technology (HKUST), Hong Kong, China, in 2018 and 2024, respectively.

He is currently a Research Associate with the Department of Electronic and Computer Engineering, HKUST. His research interests include control, perception, navigation, and path planning for autonomous robots.



Yan Ning received the B.Eng. degree in mechanical and aerospace engineering and the M.Phil. degree in electronic and computer engineering, in 2018 and 2023, respectively, from the Hong Kong University of Science and Technology (HKUST), Hong Kong, China, where he is currently working toward the Ph.D. degree with the Department of Electronic and Computer Engineering.

His research interests include hardware design, model-based and learning-based locomotion control, and planning for legged robots.

Authorized licensed use limited to: Technion Israel Institute of Technology. Downloaded on September 11,2026 at 02:01:38 UTC from IEEE Xplore. Restrictions apply.



Hongming Chen (Graduate Student Member, IEEE) received a B.Eng. degree in software engineering from the University of Electronic Science and Technology of China, Chengdu, China, in 2024. He is currently working toward the M.Phil. degree (under the supervision of Prof. Ximin Lyu) with the Department of Intelligent Systems Engineering, Sun Yat-sen University, Shenzhen, China.

His research interests include unmanned aerial vehicles, semantic navigation, and aerial manipulation.



Peize Liu received the B.Eng. degree in software engineering from the University of Electronic Science and Technology of China, Chengdu, China, in 2022. He is currently working toward the Ph.D. degree (under the supervision of Prof. Shaojie Shen) with the Department of Electronic and Computer Engineering, Hong Kong University of Science and Technology, Hong Kong.

His research interests include unmanned aerial vehicles, state estimation in aerial swarms, and swarm systems.



Yang Xu (Graduate Student Member, IEEE) received the B.Eng. degree in software engineering from the University of Electronic Science and Technology of China, Chengdu, China, in 2023. He is currently working toward the M.Phil. degree (under the supervision of Prof. Shaojie Shen and Prof. Huan Yin) with the Division of Emerging Interdisciplinary Areas, Hong Kong University of Science and Technology (HKUST), Hong Kong SAR, China.

His research interests include unmanned aerial vehicles, sensor fusion, and robot learning.



Hao Xu received the B.Sc. degree in physics from the University of Science and Technology of China, Hefei, China, in 2016, and the Ph.D. degree in electronic and computer engineering from the Hong Kong University of Science and Technology (HKUST), Hong Kong, in 2023.

He was an Algorithm Engineer with Shenzhen DJI Sciences and Technologies Ltd. He is currently an Associate Professor with the Institute of Unmanned Systems, Beihang University, Beijing, China. His research interests include unmanned aerial vehicles, aerial swarms, state estimation, sensor fusion, localization, and mapping.



Ximin Lyu received B.Eng. and M.Phil. degrees in aircraft manufacturing from the Harbin Institute of Technology, Harbin, China, in 2012 and 2014, respectively, and the Ph.D. degree in electronic and computer engineering from the Hong Kong University of Science and Technology (HKUST), Hong Kong, China, in 2019.

In 2018, he was a Senior Flight Control Researcher with Shenzhen DJI Sciences and Technologies Ltd., China. Since 2021, he has been an Associate Professor with the Department of Intelligent Systems Engineering, Sun Yat-sen University, Shenzhen, China. His research interests include robotics and controls, with a specific focus on UAV/UGV design, control, and planning.



Shaojie Shen received the B.Eng. degree in electronic engineering from the Hong Kong University of Science and Technology (HKUST), Hong Kong, in 2009, the M.S. degree in robotics and the Ph.D. degree in electrical and systems engineering from the University of Pennsylvania, Philadelphia, PA, USA, in 2011 and 2014, respectively.

He is currently an Associate Professor with the Department of Electronic and Computer Engineering and the Founding Director of the HKUST-DJI Joint Innovation Laboratory (HDJI Lab) at HKUST. His research interests include the areas of robotics and unmanned aerial vehicles, with focus on state estimation, sensor fusion, localization and mapping, and autonomous navigation in complex environments.

Dr. Shen and his research team were the recipients of the 2023 IEEE T-RO King-Sun Fu Memorial Best Paper Award and the 2023 IEEE RA-L Best Paper Award, and the Best Student Paper Award in IROS 2018. He also achieved the Honorable Mention status for the IEEE T-RO Best Paper Award in 2018 and 2020. In addition, he was recognized as the AI 2000 Most Influential Scholar Award Honorable Mention in 2020 and consecutively from 2021-2024. He is currently a Senior Editor for ICRA 2024–2026, and an Associate Editor for IJRR 2023–2024.

Authorized licensed use limited to: Technion Israel Institute of Technology. Downloaded on September 11,2026 at 02:01:38 UTC from IEEE Xplore. Restrictions apply.